How LLMs Feel About Applicants: Capturing Social Intuition as Vectors

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Abstract

Recent interpretability work shows that emotions can be represented as linear directions in a language model's residual stream. We adapt this approach to social perception across nine open-weight models from the Gemma, Qwen, and Llama families. Using synthetic stories, we derive warmth and competence directions and then test whether name-level projections align with human ratings and whether steering these directions changes callback recommendations. Human alignment was model-dependent: both dimensions aligned positively in Gemma-3 and Qwen3.6, whereas Llama-3.1 and Qwen3-14B inverted warmth, and Gemma-4 showed mixed alignment across checkpoints. Hiring steering was similarly heterogeneous: positive interventions increased callback margins in some checkpoints, decreased them in others, and produced nonlinear or decision-stable responses elsewhere. Thus, a common extraction procedure recovers warmth and competence directions across models, but their correspondence with human perceptions and causal influence on hiring decisions are not uniform.

Code and data: `https://github.com/JorgeLastraCerda/social-intuition-vectors`

Introduction

Roughly half of United States organizations now report using artificial intelligence somewhere in recruiting [22]. Judgments that were once made by a person reading an application are increasingly produced, or at least shaped, by systems whose reasoning cannot be inspected from the outside. The question of whether those systems treat applicants evenly is no longer a hypothetical one. Existing evidence says they often do not.

Correspondence studies established decades ago that otherwise identical applications receive different responses depending on the demographic signal a name carries [3], and behavioral audits of language models report the same pattern in model-generated callback recommendations [1, 2, 10]. These audits share a design: vary an input, observe an output, measure the gap. That design establishes that a disparity exists. It does not identify what inside the model produces it.

Interpretability research offers a way to ask the second question. Abstract concepts appear to be carried in a language model's residual stream as approximately linear directions, which can be read out with a probe and altered by adding the direction back during a forward pass. Emotion concepts have been shown to behave this way [23]. Social perception is a plausible next candidate, since the psychological literature treats warmth and competence as two recurring dimensions along which people evaluate each other [9], and those same two dimensions predict callback rates in human hiring experiments [11].

We test whether warmth and competence are encoded this way, and whether the encoding matters for hiring recommendations. Using 200 short stories written to vary one dimension at a time, with a nameless and gender-neutral protagonist throughout, we build warmth and competence directions from the residual stream of nine open-weight checkpoints drawn from the Gemma, Qwen, and Llama families. Three questions follow. Do the recovered directions survive validation rather than reflecting the stories used to build them? Do they correspond to how people actually rate the applicant names in a human benchmark? Does pushing them change the model's own callback recommendation?

The answers are not uniform, and that turns out to be the substantive result. Extraction succeeds everywhere: all nine checkpoints yield directions that separate held-out stories, stand well outside a random-direction null, and reconstruct stably from disjoint halves of the corpus. Beyond that point the models diverge. Competence tracks human ratings in seven of the nine checkpoints, while warmth correlates negatively or insignificantly in four. Steering shifts the stated concept judgment everywhere, but its effect on callback recommendations varies in size, in linearity, and in some checkpoints in sign, reversing between moderate and larger interventions. Whether the representation carries a demographic signal into the decision at all depends on the model.

The contribution is primarily one of connection. Recent work has already probed warmth and competence in language-model activations [6], while representational audits in mortgage underwriting have used activation steering to expose latent demographic signals [25]. We combine elements that these lines of work study separately: mean-difference directions for the two Stereotype Content Model dimensions, causal steering, human ratings and published callback rates for the same applicant names, and a comparison across nine open-weight checkpoints. This design tests whether a representational result replicates across models and whether it connects to a matched behavioral and human benchmark.

The paper proceeds as follows. The next section describes how concepts are read out of a language model and why warmth and competence are the constructs we pursue. Methods documents the story corpus, the extraction and validation procedure, the steering interventions, and the benchmark comparison. Results reports vector validation, concept-level steering, alignment with human ratings, hiring steering, and group-level disparity and mediation, in that order. The Discussion considers what the divergence between models implies for interpreting and auditing representations of this kind, together with the limitations that bound these conclusions.

Background: Reading a Concept Out of a Language Model

From Layers to Directions. A large language model processes a piece of text by passing it through a long stack of layers, and each layer reads from and writes back to a shared running representation known as the residual stream [27, 7]. Think of the residual stream as a notebook passed from one layer to the next: every layer reads what earlier layers have written, adds its own notes, and hands the notebook onward. At any chosen depth, this running state is simply a long list of numbers, a high-dimensional vector, and circuit-level analyses have shown that specific features of the input, from low-level syntax to high-level semantic content, leave consistent geometric traces in that vector across many different contexts [19]. The idea that such traces take the form of *directions* rather than isolated coordinates goes back to word-embedding analogies, where arithmetic on word vectors captures semantic relationships [16], and the *linear representation hypothesis* extends this claim to modern transformer models: a high-level feature corresponds to a direction in the residual stream, and projecting the model's state onto that direction yields a scalar measuring how strongly the feature is present [21]. Extracting a direction for some target concept is then a matter of contrast: we collect residual-stream states from texts that express the concept (condition A) and from matched texts that do not (condition B), and take the mean difference between the two groups,

$$v = \bar{h}_A - \bar{h}_B. \tag{1}$$

Normalizing v to a unit vector $\hat{v}$ gives a concept direction, and the concept score of any new piece of text is the projection $\hat{v}^\top h$. The claim worth testing, and the one we validate here through cross-validated classification and topic-holdout accuracy, is that a direction built this way generalizes beyond the handful of texts used to construct it.

Emotion Vectors. A direction built purely from the mean-difference of Eq. 1 only tells us that high- and low-condition texts sit in different places; it does not yet tell us whether the model actually relies on that direction when it acts. Testing that requires an intervention rather than an observation: at a chosen layer, we can override the model's own internal state during inference,

$$h' = h + \alpha\,\hat{v}, \tag{2}$$

where α is a signed scalar that sets how strongly, and in which direction, we push. If the model's downstream behavior then shifts systematically as α changes, while an unrelated, orthogonal direction produces no comparable shift, the direction is not merely correlated with the concept but participates causally in the model's computation. Activation addition [26] and representation engineering [30] established this intervene-and-observe recipe and demonstrated it across a range of models and behaviors. Most recently, Sofroniew et al. [23] applied exactly this recipe to emotion concepts in Claude Sonnet: building contrastive text pairs for each emotion, they extracted mean-difference directions in the residual stream, pushed the model's state along those directions during inference, and observed systematic, interpretable shifts in the model's behavior. Their study also introduced a neutral-corpus PCA denoising procedure that strips out general evaluative valence from a direction vector, a step we adopt to separate warmth and competence content from shared narrative polarity (detailed in the Supplementary Materials). Fig. 1 illustrates the resulting picture: each emotion occupies its own direction in residual-stream space, with a numeric representation that can be extracted, inspected, and used to steer the model.

Why Warmth and Competence. Nothing in this read-and-steer recipe is specific to emotion. Any construct that is linearly encoded in the residual stream should, in principle, be extractable by the same mean-difference procedure and testable by the same steering intervention, whether that construct is an emotional state or a dimension of social perception. The Stereotype Content Model proposes that social perception organizes along two fundamental axes: warmth, capturing

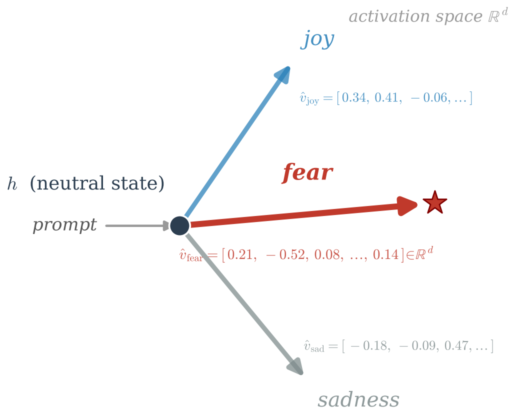

Figure 1: Emotion vectors as directions in activation space (schematic). Conceptually, an emotion vector starts from a neutral model state h. A prompt that evokes a given emotion, such as fear, pushes the residual stream toward a direction $\hat{v}_{\text{fear}}$, encoded here with its representative d-dimensional coordinates. This direction behaves like a commuter who takes the same route to work every day: the attitude, direction, and flow stay the same each time. Across thousands of different prompts that express the same emotion in different words, the model converges on this same direction through its layers, and it produces its output by moving along that direction. After Sofroniew et al. [23], who call this an emotion vector; we reuse the geometry, not the construct.

perceived intentions and trustworthiness, and competence, capturing perceived ability and effectiveness [9]. Evidence across cultural contexts suggests that warmth and competence are recurring dimensions of both interpersonal impressions and perceptions of social groups [8, 5]. These two axes therefore give us well-established, independently motivated targets to search for inside a language model, rather than an ad hoc pair invented for this study. Deas and McKeown [6] provide the closest representational precedent: they fit linear probes for the same two Stereotype Content Model dimensions and show that the decoded impressions predict response properties. Our study uses contrastive mean-difference directions rather than fitted impression classifiers, intervenes on those directions, and validates them against same-name human ratings and callback outcomes.

From Emotions to Hiring. Hiring is a domain where such a construct would matter immediately. Correspondence studies, in which otherwise identical applications differ only in a demographic signal, have established callback penalties across name-signaled ethnicity [3, 20], motherhood [4], and age [18]. Behavioral audits have already documented that large language models recommend callbacks unevenly across demographic groups [1, 2, 10], yet none of that work locates the representation inside the model that might be doing the work. In a neighboring high-stakes setting, Tripathy and Buckmann [25] identify racially associated internal directions in mortgage-underwriting prompts and show through activation steering that these representations can causally alter decisions even when baseline outputs appear fair. Their intervention targets demographic mean differences; ours targets warmth and competence and asks whether those social-perception dimensions connect demographic signals to hiring recommendations. Gallo et al. [11] supply the bridge between the two axes and these labor-market outcomes: a meta-analysis of correspondence studies of the kind just described, released together with 16,557 warmth and competence judgments from 787 raters covering 282 applicant first names drawn from nine source studies. Across the eight name-based studies with usable callback data, a more favorable first principal component combining warmth and competence was moderately associated with higher callback rates, though the effect varied substantially across studies and its prediction interval included negative relationships. That dataset serves two purposes here, as an external validity check on our probes and as the reference point against which model-level disparities are compared. We therefore ask whether a social-perception construct known to predict human callback outcomes is encoded the way emotion concepts are, and whether steering it changes a model's own callback recommendations.

Methods

Story Preparation. Probing warmth and competence directly from hiring prompts would confound the social-perception dimensions with the hiring decision itself, so the direction vectors come from a separate set of non-hiring materials. Identity control matters here. In unconstrained pilot generations, the language model systematically assigned male, Anglo-named protagonists to low-warmth and low-competence stories, which would have introduced demographic expectations before the hiring evaluation began. The corpus contains 200 narratives of approximately 90–150 words, divided evenly among high warmth, low warmth, high competence, and low competence (50 stories per condition). Its 50 everyday topics span ten domains, including workplace, learning, community, sport, and travel; each topic appears once in every condition to balance situational vocabulary across poles (Tab. S.1 lists every topic with its word-count range). Every protagonist is nameless, gender-neutral, and referred to only as “they,” so the vectors reflect depicted behavior rather than the demographic signal tested later. A large language model generated the stories, which were then checked automatically for forbidden vocabulary, length balance, and demographic neutrality. Tab. S.2 records the generation and system prompt supplied to that model. For a concrete sense of the corpus, Tab. S.3 presents one story from each of the four conditions on a single shared topic, with a link to the complete set of 200 texts.

Building the Warmth and Competence Vectors. We extracted residual-stream activations from nine open-weights instruction-tuned models spanning three model families and two version generations each for Gemma and Qwen: Gemma-3-12B-it and Gemma-3-27B-it [12], Gemma-4-12B, Gemma-4-26B-A4B, and Gemma-4-31B; Qwen3-14B [29], Qwen3.6-27B, and Qwen3.6-35B-A3B; and Llama-3.1-8B-Instruct [14]. Activations were read with TransformerLens [17] for the Gemma-3, Qwen3-14B, and Llama-3.1 checkpoints, with its Bridge interface for the three Gemma-4 checkpoints, and with a native Hugging Face backend for the two Qwen3.6 checkpoints, which the released TransformerLens build does not yet support. The *Backend* column of Tab. 1 lists the tool used for each checkpoint. Sofroniew et al. [23], whose emotion-vector method motivates our approach, describe a progression from token-level emotional connotations in early layers to context-integrated representations in middle-to-late layers. They report most analyses at a layer approximately two thirds through the model, where their evidence indicates that emotion representations encode abstract content relevant to upcoming token predictions. Guided by this result, we operationalized the same relative depth as probe_layer_frac = 0.66 and fixed it across models before conducting the layer sweeps, rather than selecting a model-specific optimum. Integer rounding placed the probe between 63 and 66 percent of model depth; Tab. 1 reports the exact layer index, model width, and mean residual-stream norm for each model. For each story, activations were mean-pooled across token positions from position 50 to the end of the story, giving one d_{model}-dimensional vector per story. This offset was fixed in advance and applied uniformly across models rather than tuned per architecture: the earliest token positions carry input-agnostic outlier activations that would otherwise dominate the pooled mean [24, 28], and because the concept stories ran roughly 100 to 170 tokens, starting at position 50 discards about the first third of each narrative so that the pooled state reflects text where the warmth or competence content is already established. High- and low-condition stories cluster in distinct regions of this space (Fig. 2). The warmth and competence direction vectors are simply the mean difference between the two clusters:

$$v_W = \bar{x}_{\text{high warmth}} - \bar{x}_{\text{low warmth}},$$
$$v_C = \bar{x}_{\text{high comp.}} - \bar{x}_{\text{low comp.}}$$

where $\bar{x}_{\text{high warmth}}$ is the average pooled activation across the 50 high-warmth stories, $\bar{x}_{\text{low warmth}}$ is the same average across the 50 low-warmth stories, and v_W (respectively v_C for competence) is the resulting direction vector. The high and low conditions are a requirement of this construction rather than an incidental design choice: a mean-difference direction can only be recovered from a contrast, so each concept needs an opposing pair of story sets to define which way its vector points. A single graded warmth or competence scale would leave nothing to subtract. All nine models used the same stories, pooling rule, and prespecified seed (20260527), enabling comparison across architectures.

Model	d_{model}	Probe layer	$\lVert\bar{h}\rVert$	Backend
Gemma-3-12B-it	3840	31/48	79,722	TL
Gemma-3-27B-it	5376	40/62	61,576	TL
Gemma-4-12B	3840	31/48	97.8	Bridge
Gemma-4-26B-A4B	2816	19/30	68.4	Bridge
Gemma-4-31B	5376	39/60	188.5	Bridge
Qwen3-14B	5120	26/40	206.6	TL
Qwen3.6-27B	5120	42/64	71.9	HF
Qwen3.6-35B-A3B	2048	26/40	3.6	HF
Llama-3.1-8B-Instruct	4096	20/32	11.4	TL

Table 1: Model architecture and activation scale for nine checkpoints. d_{model} is the number of activation values carried at each token, or the representation width. Probe layer gives the zero-indexed block used to build the directions and the total blocks; 31/48 means index 31 of a 48-block model. $\lVert\bar{h}\rVert$ is the typical activation size there: for each of 200 stories, token activations are averaged and the resulting vector's length is measured; the table reports the mean of these lengths. This value calibrates steering to each model's own scale, so larger values do not imply better or stronger models. Backend is the tool used to read activations: TL is TransformerLens, Bridge is its Hugging Face Bridge interface, and HF a native Hugging Face backend used for the Qwen3.6 checkpoints that TransformerLens does not yet support.

We check that each direction is a genuine, generalizable signal rather than an artifact of the specific stories used to build it, using seven complementary checks. Five apply within a single model: 5-fold cross-validated accuracy, topic-holdout accuracy, Cohen's d against a 1,000-direction random null, split-half cosine stability, and cross-axis classification. Two compare across models: pairwise Spearman agreement on story rankings, and neutral-corpus PCA denoising. Cross-validated and topic-holdout accuracy reach 1.00 for every model on both axes. Cohen's d exceeds all 1,000 random directions in every case. Split-half cosine stays between 0.69 and 0.90, with competence more stable than warmth in every model. Cross-axis accuracy runs from 0.82 to 1.00, indicating the two axes share substantial evaluative content. All thirty-six model pairs agree positively on both axes. Denoising lowers the warmth-competence cosine in every model without eliminating it. The Probe Validation Checks section of the Supplementary Materials states what each check tests and why. Tab. 2 reports Cohen's d, the random-null z-score, split-half cosine, and calibrated cross-axis accuracy for all nine models; 5-fold cross-validated and topic-holdout accuracy reach 1.00 for every model on both axes and are reported in Results rather than repeated in the table. Two further analyses look beyond a single model's own directions:

- **Cross-model Spearman agreement**: Nine models, trained independently by different organizations, could each encode warmth as their own unrelated pattern, with no reason for their story rankings to agree with one another. To test this, every pair of the nine models is compared: each story is projected onto both models' own directions to give one score per model, and the two resulting sets of scores are correlated with Spearman's rank correlation. All thirty-six pairs show positive agreement on both axes, which means the nine architectures are converging on a shared cross-architecture construct rather than nine independent, unrelated ones.
- **Neutral-corpus PCA denoising**: Some of what warmth and competence have in common in the model has nothing to do with either concept specifically; it looks more like a general tone the model applies regardless of topic. To find and remove that general tone, we assembled 1,500 length-matched introductory passages drawn from distinct Wikipedia articles on ordinary, unrelated topics, with nothing to do with warmth or competence, and ran them through the same model and looked at what the warmth and competence directions have in common with those neutral texts. That shared, non-specific part is then subtracted out of both direction vectors, leaving only what is actually specific to warmth and to competence. The full nine-model procedure, run as a robustness check, is in Tab. S.4.

All 36 pairwise agreement scores are in Tab. S.5; Results summarizes the same comparison in a compact dot plot (Fig. 5).

Steering the Concept Vectors. Finding a direction that separates high- and low-condition stories does not yet prove the model actually uses that direction when forming its judgment. To test that, we push the model's internal activity along the direction while it answers, using activation steering [26] via TransformerLens hooks at the probe layer (Fig. 2), and see whether its answer changes.

Because different models' internal activations are different sizes (see the norm column in Tab. 1), we measure how hard we push relative to each model's own typical activation size, not in raw units, so the same push strength means the same thing across all nine models. The push itself is

$$h' = h + \alpha \, \|\bar{h}\| \, \hat{v},$$

where h is the story's original activation, $\hat{v}$ is the unit-normalized target direction, $\|\bar{h}\|$ is that model's mean residual-stream norm from Tab. 1, and α is the strength we choose.

At the concept level, the model answers a direct yes-or-no question: is this protagonist high in warmth (or competence)? Five strengths are tested, $\alpha \in \{-0.10, -0.05, 0, +0.05, +0.10\}$, spanning a strong pull toward "low" to a strong pull toward "high." This range was fixed after an initial sweep that also tested the wider strengths $\alpha \in \{\pm 0.25, \pm 0.50\}$ on Gemma-3-12B and Gemma-3-27B, where the strong settings saturated the response instead of shifting it cleanly: at ± 0.50 the model stops tracking the story and returns Yes or No almost mechanically, the same breakdown at high steering strength that Turner et al. [26] report for activation steering more generally. Saturation here follows from the structure of the concept prompt, whose answer depends directly on the story the model is asked to read, so a push strong enough to drown out that story yields a fixed answer rather than a graded one. Because that behavior is a property of the task and not of any single checkpoint, we did not repeat the wide sweep on the other seven models and applied the narrow range uniformly to all nine. The full saturation curves are reported in the Supplementary Materials. For each strength, the direction is applied to 10 held-out topics it was never built from, separate from the 40 topics used to build it. For the Gemma-4 and Qwen3.6 checkpoints this yes-or-no question is presented through each model's native chat template, since these more recent chat-tuned checkpoints do not reliably return a clean single-token Yes or No to a raw-text prompt; the other four models receive it as raw text, matching how each checkpoint was run throughout.

We record how much more the model favors Yes over No at each strength, the next-token logit margin:

$$\Delta = \mathrm{logit}(\mathrm{Yes}) - \mathrm{logit}(\mathrm{No}),$$

where logit(Yes) and logit(No) are the model's raw output scores for the two answer tokens before any softmax normalization, so a positive Δ means the model favors Yes and a larger Δ means a stronger preference. Throughout, the raw dense directions define the primary causal comparisons reported here; the PCA-denoised vectors serve as a robustness check (Supplementary, PCA Denoising).

A single steering result does not tell us whether the effect is specific to warmth or competence, or whether it would show up for almost any direction we pushed along. To check this, we compare six different directions and see how each one shifts the model's answer:

- **Dense target direction**: The same warmth or competence direction used everywhere else in this study, the difference between the average high- and low-condition activations. This is the main direction we are actually trying to validate.

- **GS2 Decoded Gemma Scope 2 SAE direction**: Gemma Scope 2 [13] is a separate tool from Google DeepMind that breaks a model's internal activity down into thousands of smaller, more specific pieces. We compute the same warmth/competence contrast inside this alternative breakdown, then translate it back into the model's own activation space, to see whether steering still works when built from a completely different representation.
- **GS2 Axis-specific component**: Within that breakdown, some pieces move together for both warmth and competence, and some move only for whichever concept is being tested. This direction keeps only the part specific to the one concept being tested.
- **GS2 Component shared between the two axes**: The opposite of the direction above, only the part that moves for both warmth and competence together, the piece that is not specific to either concept on its own.
- **GS2 Opposing concept axis**: When testing warmth, we steer along the competence direction instead, and vice versa when testing competence, to see whether an unrelated concept's direction moves the answer too.
- **Random orthogonal control**: A direction chosen at random and adjusted so it does not overlap at all with the real target direction, our chance-level baseline.

The dense target direction and a random orthogonal control were both tested on all nine models; the other four were tested only on Gemma-3-12B and Gemma-3-27B. Those four, marked GS2 above, are built from a Gemma Scope 2 decomposition, and that tool covers only the Gemma-3 generation [13], so they could not be extended to Gemma-4, Qwen, or Llama. The nine-model random control is a differently constructed, calibrated set of many random directions per model rather than the single orthogonalized vector compared here, but it serves the same chance-level baseline role. If steering only shifts the answer along the real target direction, and not along the unrelated ones, that supports the effect being specific to warmth or competence rather than an artifact of pushing the residual stream in any direction. Uncertainty throughout is estimated by paired topic bootstrap resampling.

Steering Hiring Decisions. The concept-level test shows the model will move its stated judgment when pushed; whether that carries over to an actual hiring decision is a separate question. To answer it, we bring the callback task into the same steering setup. The hiring prompt pairs each of the 282 applicant first names from Gallo et al. [11] with one fixed administrative-assistant application, identical in education, experience, and skills across every trial, so the name is the only signal that changes. The model reads the application and answers

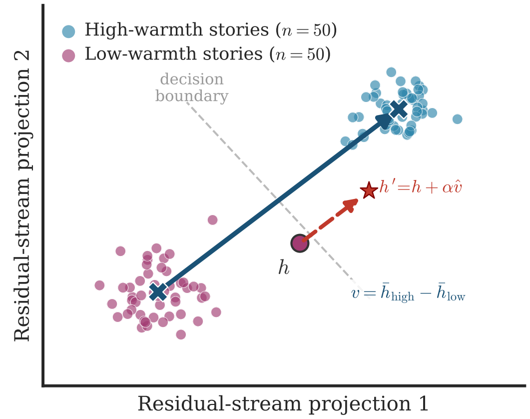

Figure 2: Residual-stream concept geometry (schematic). Low-warmth and high-warmth stories occupy different regions of this space, and the mean-difference direction $\boldsymbol{v} = \bar{\boldsymbol{h}}_{\mathbf{high}} - \bar{\boldsymbol{h}}_{\mathbf{low}}$ (blue arrow) is what separates them. This panel shows only the warmth axis, so its v is the vector written v_W in Methods, with competence v_C defined the same way. Steering acts like grabbing the wheel of a car that is already driving down one street and turning it until the car crosses onto the next street over: the red dashed arrow, $\boldsymbol{h'} = \boldsymbol{h} + \boldsymbol{\alpha}\,\hat{\boldsymbol{v}}$, shows this same kind of forced redirection, pushing a story's representation from the low-warmth region, across the decision boundary, into the high-warmth region. Concretely, this is done by adding a scaled copy of that same high-warmth direction to the story's current representation. Illustrative synthetic activations, not measured data.

a single question, "would you recommend calling this candidate back for an interview? Answer with a single word: Yes or No." We score that answer with the callback margin,

$$\Delta_{\text{callback}} = \text{logit}(\text{Yes}) - \text{logit}(\text{No}),$$

the same Yes-versus-No logit difference used at the concept level, now read from the hiring prompt rather than the concept-judgment prompt. Demographic labels stay out of the prompt. Race and gender are never shown to the model; each name is tagged afterward from the Gallo et al. [11] coding, mirroring how human correspondence studies read group membership from names alone [3]. We then apply the same steering hooks used at the concept level while the model works through these applications, over a 60-name subset of the roster. Strength is swept broadly at $\alpha \in \{\pm 0.25, \pm 0.50\}$ and refined at $\{\pm 0.05, \pm 0.10\}$ on all nine models, so that a push on the warmth or competence direction can be traced directly to a shift in the callback margin. No high-versus-low split is applied to the names at this stage either; the intervention lives entirely in the signed strength α, which varies continuously, while each name only fixes which application the model reads. Unlike the concept judgment, the hiring decision does not saturate under the wider strengths, so both regimes stay usable here.

Its callback answer turns on a single applicant name set against a fixed, neutral application, and the model already leans one way before any push, so steering at ± 0.50 scales that existing leaning up or down rather than locking the answer, and the response stays close to linear across the broad grid for roughly a third of model-axis combinations, though several checkpoints depart from it (Results reports the per-model fit). Refining to ± 0.05 and ± 0.10 then resolves the smaller, near-baseline shifts that the wider grid steps over. Per-model steering curves and slopes are reported in the Results.

Names, Human Ratings, and Disparity. Beyond the causal steering test, we ask whether the underlying representations track real human perception and real-world demographic gaps, using four analyses:

- **Probe-versus-human alignment**: Independently of the hiring prompt, each applicant name is embedded in the neutral carrier sentence "The job applicant's name is {Name}.", and the resulting residual-stream activation is projected onto the raw warmth and competence directions to yield a per-name probe score; no denoised counterpart is computed at this stage. We compare these scores against the aggregated human ratings behind them (16,557 rater judgments from 787 raters across nine source studies) using Spearman rank correlation, which tells us whether the model's internal sense of a name's warmth or competence lines up with how people actually perceive that name. Each name already carries a continuous crowdsourced warmth and competence score, so no high-versus-low grouping is imposed here, unlike the concept stories used to build the directions. The correlation runs directly on the graded ratings, since sorting names into poles would discard information the continuous scale already carries.
- **Group-level callback disparity**: Callback disparities are computed as differences in mean callback margin between name groups (race: 47 Black-signaling versus 180 White-signaling names; gender: 154 female versus 115 male). Each model difference is divided by the pooled within-group standard deviation of its callback margin. The corresponding human difference uses the same matched-name roster and the pooled within-group standard deviation of the published callback rate, which puts both outcomes on the same standardized mean-difference scale. For the Gemma-4 and Qwen3.6 checkpoints the hiring prompt is presented through each model's native chat template, since these more recent chat-tuned checkpoints do not reliably return a clean single-token Yes or No to a raw-text prompt; the other five models receive it as raw text.
- **Human reference gaps**: The main standardized comparison averages the published callback observations to one rate per first name before matching, so each matched name receives one observation and the human gap uses exactly the same roster as the corresponding model gap. The study-specific raw companion analysis retains one observation per name-study pair instead. Some source studies (Neumark, Farber) also vary applicant age as an experimental factor; because our name ratings carry no age dimension, that companion averages callback rates across age within each name-study pair. In total, 246 name-study observations across 186 distinct names match the study-specific benchmark.
- **Bootstrap mediation**: Finally, we test whether the internal warmth and competence representations actually carry the name-group signal into the callback decision, using bootstrap mediation along the path name group $\rightarrow$ probe score $\rightarrow$ callback margin (5,000 resamples, seed 20260527, $n = 269$ names with complete data, race subset $n = 227$), reporting indirect effects with 95% confidence intervals. Results reports this procedure for all nine models and both race and gender groupings, thirty-six combinations in total.

Results

Vector Validation, All Nine Models. Every one of the five single-model checks and two cross-model analyses described in Methods was run on all nine checkpoints, and none flags a direction as an artifact of the specific stories used to build it.

Tab. 2 summarizes three of the five single-model checks in four reported statistics. Cohen's d and the random-null z-score jointly report the random-direction separation check, followed by split-half cosine and calibrated cross-axis accuracy. The 5-fold cross-validated and topic-holdout checks are not columns in that table because both reach exactly 1.00 for every model on both axes and every fold, so there is no variation left to tabulate. Cohen's d ranges from 2.70 (Gemma-3-12B, warmth) to 9.97 (Qwen3-14B, competence), and the random-direction null comparison never comes close: across all nine models and both axes, 0 of the 1,000 random directions tested per model matched or exceeded the real direction's separation, with z-scores between 3.7 and 15.1. Split-half cosine stability stays in the 0.69-0.90 range reported in Methods, and calibrated cross-axis accuracy stays between 0.82 and 1.00. Fig. 3 shows the same overlap geometrically: the angle between the warmth and competence directions varies by model, from 41.5 degrees at Gemma-3-12B to 60.4 at Gemma-4-12B, so the two axes are neither collinear nor orthogonal in any checkpoint. Together these confirm that warmth and competence share substantial evaluative content rather than behaving as fully separate constructs.

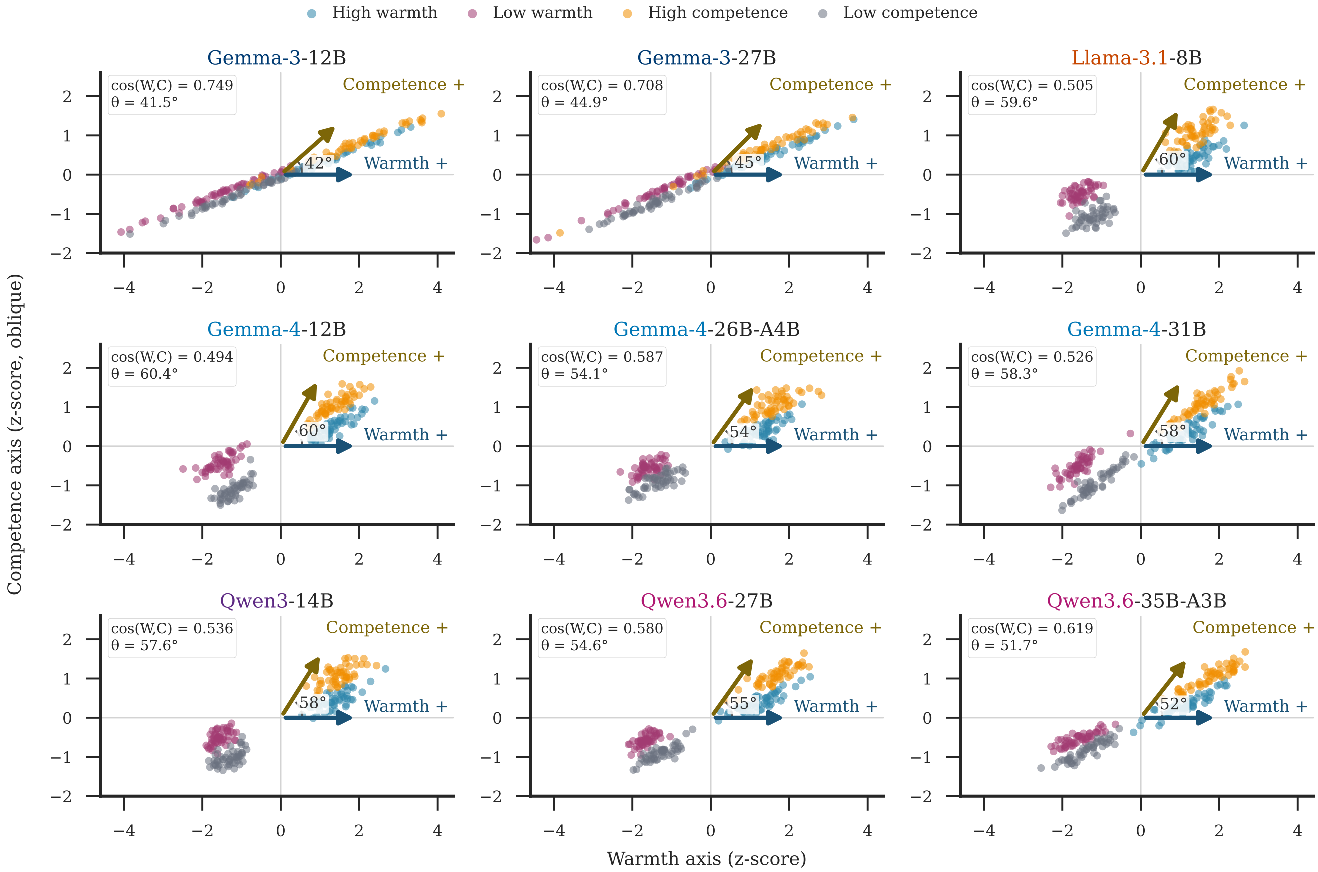

Figure 3: Warmth and competence directions share model-dependent geometry. Each panel projects the same 200 synthetic stories onto raw dense probe-layer directions in an oblique basis. The arrow angle equals $\mathbf{arccos[cos(\boldsymbol{v}_W, \boldsymbol{v}_C)]}$, so smaller angles indicate stronger geometric alignment. All panels use model-internal z-scores, shared axis limits, and equal x/y scaling.

Fig. 4 traces the same separation across depth rather than across steering strength. Cohen's d generally strengthens through the early and middle layers, while the dotted line marks the prespecified probe depth of 0.66 used for every model. This shared reference point is not a model-specific optimum and rarely coincides with peak separability, as discussed in Limitations.

Agreement across architectures is just as consistent. Spearman correlation on the 36 possible model pairs never turns negative: overall ranking agreement (which includes the high-versus-low split itself) ranges from 0.741 to 0.992 across both axes, and the harder within-condition comparison, which asks whether two models rank stories the same way inside one condition rather than just separating high from low, ranges from 0.095 to 0.899 with a median near 0.46 for warmth and 0.54 for competence. Fig. 5 shows all 36 model pairs for each summary; the complete numeric table is in Tab. S.5.

Neutral-corpus PCA denoising, reported in full in Tab. S.4, shows the same pattern across every model: removing the components shared with 1,500 length-matched introductory passages drawn from distinct Wikipedia articles lowers the warmth-competence cosine but never drives it to zero. The reduction is largest in the two Gemma-3 checkpoints (0.749 to 0.530 and 0.708 to 0.487) and comparatively small elsewhere, typically a few hundredths, suggesting that most of what remains after denoising is genuine conceptual overlap rather than a removable general tone.

Steering the Concept Vectors, All Nine Models. At the narrow local grid ($\alpha \in \{-0.10, ..., +0.10\}$), Fig. 6 tracks the held-out concept margin against steering strength in every model, normalized by each model's own baseline high-low gap so the curves stay comparable despite very different raw logit scales. Gemma-3-12B reaches the largest positive endpoint on both axes (0.236 warmth, 0.141 competence), Qwen3-14B is second on both (0.122, 0.104), and Gemma-4-26B-A4B and Gemma-4-31B stay nearly flat at the competence endpoint (-0.002 and -0.001) despite separating high-competence from low-competence stories cleanly in the

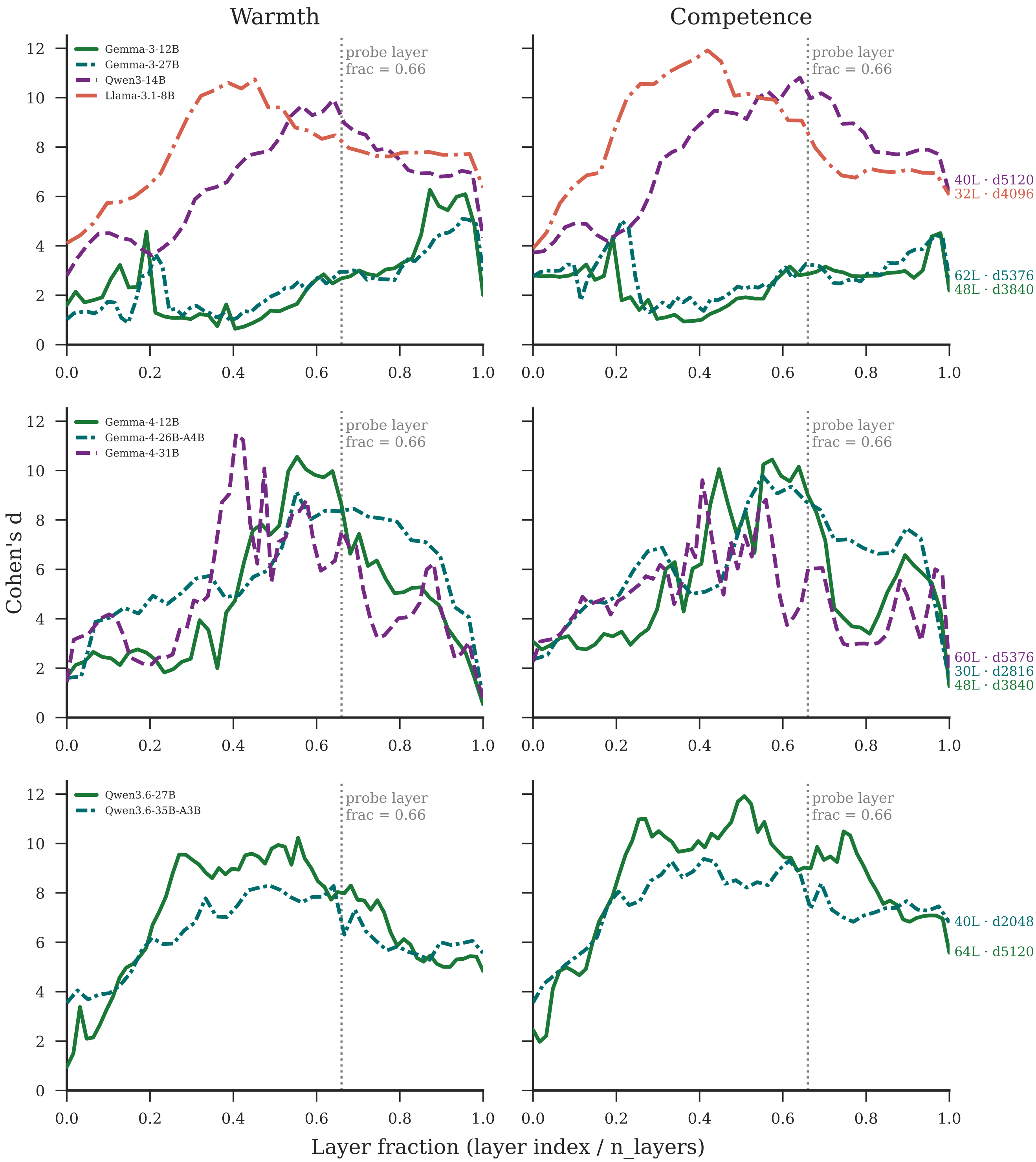

Figure 4: Warmth and competence directions emerge robustly across depth. Layer-wise Cohen's d across nine open-weights models. The dotted vertical line marks the fixed probe-layer fraction of 0.66.

checks above. A model can encode a concept strongly while still responding weakly to an additive push along its own mean-difference direction.

Wide strengths tell a partly different story. The sweep reported in Tab. S.9 tests $\alpha \in \{\pm 0.25, \pm 0.50\}$ on Gemma-3-12B and Gemma-3-27B, the two models tested at this range. Three of the four model-axis rows show the step from +0.25 to +0.50 shrinking relative to the step before it, matching the saturating breakdown Methods describes for extreme strengths; Gemma-

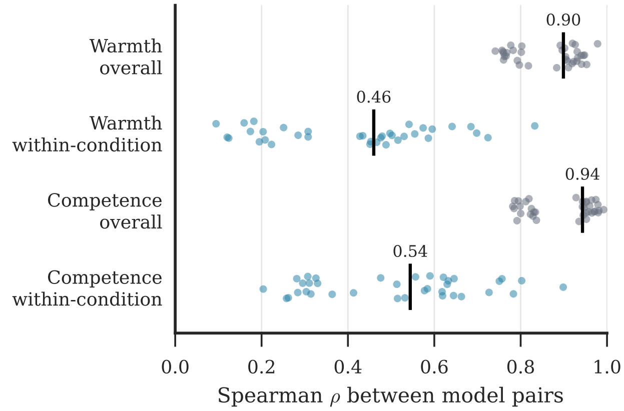

Figure 5: Cross-model story-ranking agreement, all nine models. Each dot is one of the 36 model pairs. Overall correlations include the high-versus-low split; within-condition correlations compare story ordering inside a shared condition. Vertical marks show medians. Full pairwise values are in Tab. S.5.

Model	d	z	cos	acc.
G3-12B	2.70 / 2.83	3.9 / 3.7	0.82 / 0.90	0.87 / 0.82
G3-27B	2.95 / 3.27	4.3 / 4.5	0.81 / 0.88	0.90 / 0.86
L3.1-8B	8.48 / 9.07	15.0 / 15.1	0.70 / 0.83	0.99 / 1.00
G4-12B	8.63 / 9.04	14.1 / 12.8	0.77 / 0.87	1.00 / 0.98
G4-26B	8.36 / 8.75	15.0 / 14.3	0.69 / 0.85	1.00 / 1.00
G4-31B	7.56 / 6.03	13.1 / 8.6	0.73 / 0.86	1.00 / 0.95
Q3-14B	8.97 / 9.97	14.1 / 14.6	0.71 / 0.85	1.00 / 1.00
Q3.6-27B	7.98 / 8.99	12.9 / 12.9	0.76 / 0.86	0.99 / 1.00
Q3.6-35B	6.31 / 7.35	11.3 / 10.4	0.71 / 0.86	1.00 / 0.97

Table 2: Probe validation, all nine models. Model names are abbreviated (G3, G4, L3.1, Q3, Q3.6 for the Gemma-3, Gemma-4, Llama-3.1, Qwen3 and Qwen3.6 families). Columns are Cohenś d, the random-null z, split-half cosine, and calibrated cross-axis accuracy. Warmth (W) and competence (C) values in each cell. Cohen's d is the effect size separating high- and low-condition stories on the raw dense direction. Random-null z standardizes that same d against 1,000 random directions' own high/low separation on the same stories; in every model and axis, 0 of 1,000 random directions matched or exceeded the real direction. Split-half cosine is the cosine between two directions independently built from non-overlapping 25-story halves of each condition. Cross-axis accuracy (calibrated) is how well a story's projection onto one axis's direction predicts its label on the other axis. 5-fold cross-validated and topic-holdout accuracy are omitted here since both are 1.00 for every model on both axes.

3-27B's warmth response is the exception, still accelerating at +0.50 rather than leveling off. The two-model table is retained in the Supplementary Materials because this calibration sweep is not available for the other seven checkpoints.

A single direction moving the answer does not by itself establish that the effect is specific to warmth or competence rather than any sufficiently large push. Tab. 3 compares six directions at the local endpoint on the same two models. The dense target direction produces the row's largest effect in only one of the four rows (Gemma-3-12B warmth); in the other three, an unrelated direction, most often the SAE-decoded or axis-specific component, matches or exceeds it. This does not read as clean specificity, and points instead toward the same shared evaluative content the neutral-corpus PCA denoising already found between warmth and competence. Two further checks, reported in the Supplementary Materials, extend this picture: feature ablation (Tab. S.7) shows scale-specific necessity. Removing the features shared between axes shrinks both gaps in Gemma-3-27B but increases both gaps in Gemma-3-12B, so the same shared mechanism does not hold uniformly across checkpoints. Cross-scale feature matching (Tab. S.8) finds Gemma-3-12B's and Gemma-3-27B's SAE features correlate well above a permutation null (Tab. S.6 reports the underlying reconstruction quality behind both checks).

Comparing the dense target against a random-direction control extends across all nine models, but the control itself is not built the same way everywhere. Tab. 4 reports both effects for every model and axis, with the heterogeneous control basis stated in the caption: five checkpoints (Gemma-4-12B, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.6-27B, Qwen3.6-35B-A3B) carry a calibrated set of 99 SD-matched random directions, while the four original checkpoints carry a single random direction. Where the calibrated control exists, the result splits by family: both Qwen3.6 checkpoints place the dense-target effect clearly outside the random control's interval, while all three Gemma-4 checkpoints have a random interval wide enough to contain the dense-target effect, so this comparison alone does not establish specificity for that family. The calibrated 99-direction control is not available for the four legacy checkpoints, so the comparison there rests on a single

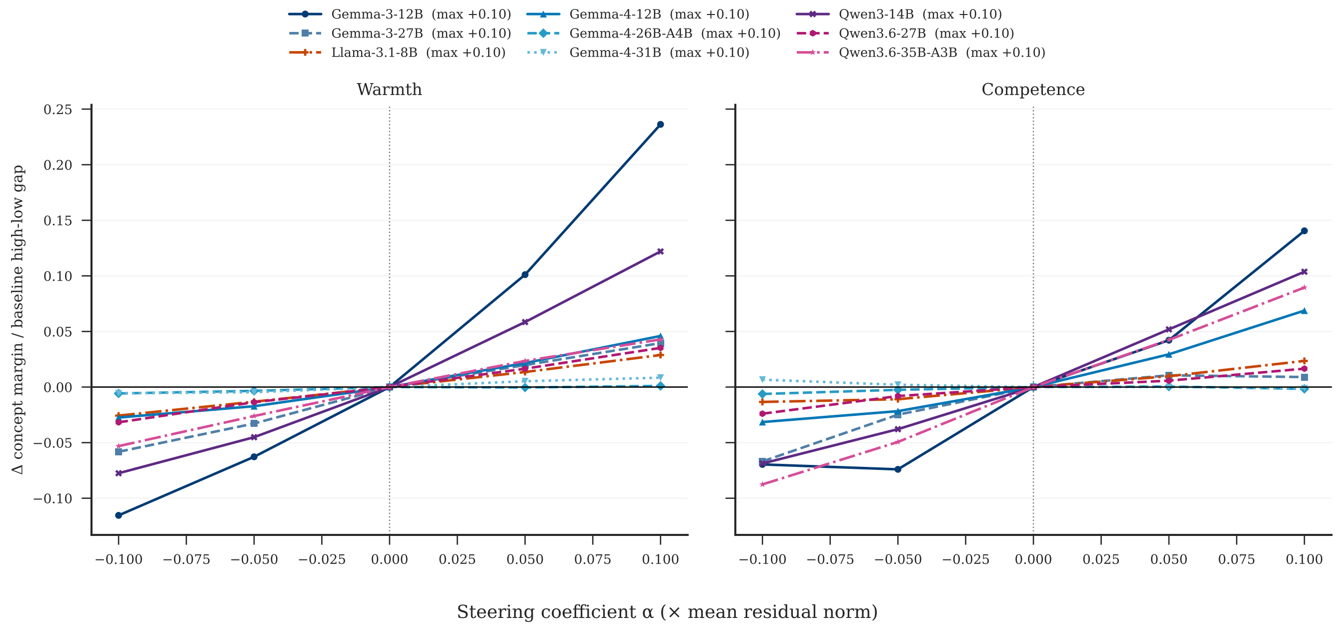

Figure 6: Normalized concept-level steerability varies across nine checkpoints. Each curve shows the change in the held-out concept margin $\mathbf{\Delta}$ (the Yes-versus-No logit difference defined in Methods) produced by target-direction steering, divided by the baseline high-versus-low gap in $\mathbf{\Delta}$ for the same model and axis. Parentheses in the legend report the maximum positive steering coefficient, $\alpha = +0.10$, expressed relative to the mean residual norm. Normalization reduces raw logit-scale differences but does not establish direction specificity; calibrated random and cross-axis controls are interpreted in the text.

random draw.

Model	Axis	Target	Control range	Exceeds control?
G3-12B	Warmth	+3.81	+0.10 [-0.32, +0.49]	yes
G3-12B	Competence	+1.98	+0.30 [+0.05, +0.56]	yes
G3-27B	Warmth	+1.02	+0.61 [+0.32, +0.89]	yes
G3-27B	Competence	+0.21	-3.36 [-4.00, -2.76]	yes
L3.1-8B	Warmth	+0.26	+0.06 [+0.03, +0.10]	yes
L3.1-8B	Competence	+0.18	-0.12 [-0.14, -0.11]	yes
G4-12B	Warmth	+1.84	+0.29 [-2.48, +3.07]	no
G4-12B	Competence	+2.66	+0.23 [-2.33, +2.79]	no
G4-26B	Warmth	+0.05	-0.84 [-3.79, +2.10]	no
G4-26B	Competence	-0.07	-0.66 [-3.37, +2.05]	no
G4-31B	Warmth	+0.44	+0.85 [-2.31, +4.02]	no
G4-31B	Competence	-0.06	+0.78 [-1.76, +3.32]	no
Q3-14B	Warmth	+1.20	-0.02 [-0.10, +0.06]	yes
Q3-14B	Competence	+1.11	+0.23 [+0.17, +0.28]	yes
Q3.6-27B	Warmth	+0.69	+0.01 [-0.03, +0.05]	yes
Q3.6-27B	Competence	+0.31	+0.01 [-0.02, +0.04]	yes
Q3.6-35B	Warmth	+0.82	+0.00 [-0.06, +0.07]	yes
Q3.6-35B	Competence	+1.45	-0.02 [-0.08, +0.05]	yes

Table 4: Dense-target steering versus a random-direction control, all nine models. The random-control basis is not uniform: Gemma-4-12B, Gemma-4-26B-A4B, Gemma-4-31B, Qwen3.6-27B, and Qwen3.6-35B-A3B use 99 SD-matched random directions per model, reported as mean and an approximate 95% random-direction range (mean ±1.96 sample SD); Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, and Qwen3-14B use a single random direction, reported with its own bootstrap CI. "Exceeds control?" indicates whether the target lies outside its reported control range. 12 of 18 rows exceed control; none of the six Gemma-4 rows do.

Alignment with Human Ratings, All Nine Models. Everything reported so far describes the nine models behaving alike: the directions separate the stories everywhere, and steering moves the stated concept judgment everywhere. Whether those directions correspond to human social perception is a separate question, and it is the point at which the models stop agreeing.

Independently of the hiring prompt, each of the 282 rated applicant names is embedded in the neutral carrier sentence "The job applicant's name is {Name}.", and the resulting residual-stream activation is projected onto the raw warmth and competence directions to yield a per-name probe score. Tab. 5 correlates these scores with the aggregated human ratings behind them (16,557 rater judgments from 787 raters across nine source studies) using Spearman rank correlation. Warmth alignment ranges from $\rho = -0.287$ (Llama-3.1-8B) to +0.388 (Gemma-3-27B); competence alignment ranges from $\rho = -0.058$, not significant (Llama-3.1-8B), to +0.466 (Qwen3-14B). Three models show significant negative warmth alignment: Llama-3.1-8B ($\rho = -0.287$), Gemma-4-26B-A4B ($\rho = -0.190$), and Qwen3-14B ($\rho = -0.178$). Gemma-4-12B is the fourth model without positive significant warmth alignment, with a near-zero, nonsignificant correlation ($\rho = +0.009$). Competence tracks human perception more consistently: seven correlations are positive and significant, while the remaining two are slightly negative and nonsignificant (Llama-3.1-8B, $\rho = -0.058$;

Model	Axis	Dense target	SAE decoded	Axis-specific	Shared	Opposing axis	Random
G3-12B	Warmth	**+3.88**	+2.30	+2.10	+2.16	+3.24	-0.32
G3-12B	Competence	+2.01	**+3.69**	+3.12	+2.34	+2.38	+2.88
G3-27B	Warmth	+1.02	-0.19	-1.38	+0.69	**+2.86**	+1.34
G3-27B	Competence	+0.21	+3.53	**+4.36**	+1.61	+0.31	+2.67

Table 3: The dense target is strongest in only one of four model-axis comparisons. This comparison is limited to Gemma-3-12B and Gemma-3-27B because Gemma Scope 2 is available only for these checkpoints. Cells report the change in the held-out Yes-versus-No logit margin for six directions at $\alpha = +0.10$. Dense target is the same mean-difference direction used everywhere else in this study; SAE decoded, axis-specific, and shared are built from a Gemma Scope 2 sparse decomposition; opposing axis steers along the other concept's direction; random is a single orthogonalized control. Bold marks the row's largest effect. Dense target is largest in only 1 of the four model-axis rows; the other three are matched or exceeded by at least one unrelated direction, most often the SAE-decoded or axis-specific component, which does not support a clean specificity reading and is discussed in the text alongside the shared warmth-competence overlap already noted in PCA denoising.

Gemma-4-26B-A4B, $\rho = -0.044$). The same table's callback columns report a separate quantity, the correlation between each model's own probe projection and its own unsteered callback margin, which is not a human-alignment measure.

Model	$L(f)$	$\rho_{W,H}$	$\rho_{C,H}$	$\rho_{Y,W}$	$\rho_{Y,C}$
G3-12B	31 (0.65)	+0.357***	+0.237***	+0.150*	+0.149*
G3-27B	40 (0.65)	+0.388***	+0.271***	-0.160**	-0.156**
L3.1-8B	20 (0.62)	-0.287***	-0.058 n.s.	+0.092 n.s.	+0.053 n.s.
G4-12B	31 (0.65)	+0.009 n.s.	+0.216***	-0.110 n.s.	-0.124*
G4-26B	19 (0.63)	-0.190**	-0.044 n.s.	+0.510***	+0.373***
G4-31B	39 (0.65)	+0.254***	+0.290***	+0.024 n.s.	+0.337***
Q3-14B	26 (0.65)	-0.178**	+0.466***	+0.195**	+0.426***
Q3.6-27B	42 (0.66)	+0.192**	+0.247***	+0.302***	+0.220***
Q3.6-35B	26 (0.65)	+0.206***	+0.130*	+0.344***	+0.197***

Table 5: Name-level correlation between probed warmth/competence and human ratings. For each of the 282 rated applicant names, the model's residual-stream activation at the probe layer is projected onto the warmth and competence direction vectors. "Probe layer" gives the selected layer index and its resolved fraction of total depth (targeted probe_layer_frac = 0.66; the reported fraction is the integer-rounded layer divided by model depth). Subscripts W, C, H, and Y denote warmth, competence, the corresponding human rating, and callback margin. The first two correlation columns compare probe projections with human ratings; the last two compare the probe projection with the model's own unsteered callback margin. All correlations are Spearman ρ. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$, n.s. not significant.

Steering Hiring Decisions, All Nine Models. Pushing the warmth or competence direction while the model reads a hiring application shifts the callback margin Δ_{callback} in every model, tested on the same 60-name subset at the broad strength grid $\alpha \in \{\pm 0.25, \pm 0.50\}$, chosen here uniformly across all nine models so the comparison sits on one shared scale rather than mixing the per-model preferred endpoints used by Tab. S.10.

Tab. 6 fits a line through each model-axis response and reports the slope alongside R^2. The fit is close to linear in only 7 of the 18 rows: Gemma-3-12B warmth; both Llama-3.1-8B axes; Gemma-4-26B-A4B warmth; both Qwen3.6-27B axes; and Qwen3.6-35B-A3B warmth. The remaining eleven depart from a straight line, most visibly Gemma-3-27B, where R^2 stays below 0.35 on both axes and the endpoint delta at $\alpha = +0.50$ is actually negative despite a positive slope estimate. Slopes vary in sign and size as well as fit quality: Gemma-3-12B warmth has the steepest response (+12.91 per unit α), and Qwen3-14B competence is the only row with a negative slope (-0.11), effectively flat. That broad negative fit is range-dependent: on the local grid, Qwen3-14B competence increases by +0.010 at $\alpha = +0.10$ and by +0.204 from $\alpha = -0.10$ to +0.10. The sign therefore changes with the tested range, while the local endpoint remains practically small. Fig. 7 makes the divergence concrete on matched examples: the same positive intervention at $\alpha = +0.10$ moves the mean decision from No to Yes at Gemma-3-12B and from Yes to No at Gemma-3-27B, using the same 60 applications in both panels. The categorical Yes/No transitions behind this same steering intervention, including how often it crosses the decision boundary rather than only scaling an existing lean, are reported in full in Tab. S.10.

Model	Axis	Slope	R^2	$\Delta_{+0.50}$
G3-12B	Warmth	+12.91	**0.925**	+8.35
G3-12B	Competence	+9.07	0.659	+6.23
G3-27B	Warmth	+1.16	0.029	-0.23†
G3-27B	Competence	+2.96	0.347	-0.98†
L3.1-8B	Warmth	+3.97	**0.894**	+3.17
L3.1-8B	Competence	+2.50	**0.862**	+2.17
G4-12B	Warmth	+8.76	0.654	-1.26†
G4-12B	Competence	+6.48	0.649	-0.13†
G4-26B	Warmth	+3.60	**0.905**	+0.75
G4-26B	Competence	+0.65	0.061	-1.99†
G4-31B	Warmth	+7.83	0.509	-2.59†
G4-31B	Competence	+3.07	0.209	-3.64†
Q3-14B	Warmth	+1.85	0.799	+0.60
Q3-14B	Competence	-0.11	0.066	-0.38
Q3.6-27B	Warmth	+5.69	**0.938**	+2.24
Q3.6-27B	Competence	+4.87	**0.917**	+1.07
Q3.6-35B	Warmth	+3.73	**0.909**	+1.23
Q3.6-35B	Competence	+1.45	0.296	-1.09†

Table 6: Hiring callback steering, broad grid, all nine models. Mean change in the callback margin Δ_{callback} across the 60-name subset, at $\alpha \in \{\pm 0.25, \pm 0.50\}$. Slope and R^2 come from an ordinary least-squares fit of the five per-strength means; bold marks $R^2 \geq 0.800$. A dagger marks an endpoint whose sign disagrees with the fitted slope (8 of 18 rows). The fit is close to linear in only 7 of the 18 model-axis rows; several others, most visibly Gemma-3-27B on both axes, show weak or non-monotonic trends instead.

Group-Level Disparity and Mediation, All Nine Models. Two further analyses move from individual names to demographic groups, and from correlation to the stronger question of whether the representations carry a group signal into the callback decision.

Tab. 7 compares race and gender callback gaps on one standardized scale. Seven models favor Black-signaling names and two favor White-signaling names, while the pooled human race reference is small at $d = +0.15$ and is therefore better described as near zero than as a directional penalty. Gender produces the clearer mismatch: eight models favor female-signaling names, whereas the human benchmark favors male-signaling names at $d = -0.47$. Gemma-3-27B is the only checkpoint with a negative model gender gap, at $d = -0.46$. Each value divides the group mean difference by the pooled within-group standard deviation for that outcome on the exact matched-name population. The underlying standardized representation scores and raw group margins are in Tab. S.11, with study-specific values in Tab. S.12.

	Race: Black − White		Gender: Female − Male	
Model	Model d	Human d	Model d	Human d
G3-12B	-0.09	+0.15	+0.65	-0.47
G3-27B	+1.44	+0.15	-0.46	-0.47
L3.1-8B	+0.45	+0.15	+0.46	-0.47
G4-12B	+0.12	+0.15	+0.18	-0.47
G4-26B	+0.71	+0.15	+1.10	-0.47
G4-31B	+0.20	+0.15	+0.36	-0.47
Q3-14B	+0.16	+0.15	+0.98	-0.47
Q3.6-27B	+0.29	+0.15	+0.40	-0.47
Q3.6-35B	-0.23	+0.15	+1.36	-0.47

Table 7: Model and human callback gaps by demographic axis. Every entry is a standardized mean difference, d, computed as the positive-group mean minus the negative-group mean divided by that outcome's pooled within-group SD on the exact matched-name population. Race compares 47 Black-signaling with 180 White-signaling names; gender compares 154 female-signaling with 115 male-signaling names. Human values repeat because every model uses the same benchmark and name roster. Positive race values favor Black-signaling names; positive gender values favor female-signaling names. Underlying group levels are reported in Tab. S.11.

The last analysis asks whether the probed representations actually carry the name-group signal into the callback decision, rather than merely correlating with it. Bootstrap Mediation, All Nine Models in the Supplementary Materials tests this with bootstrap mediation along the path name group → probe score → callback margin, run separately for race and gender and for warmth and competence, thirty-six tests across the nine models. Fourteen of the thirty-six indirect effects have unadjusted 95% bootstrap intervals that exclude zero. No multiplicity-adjusted inference is reported, so all path comparisons are exploratory. At this unadjusted threshold, the two Gemma-3 checkpoints have no interval excluding zero on either grouping or probe, consistent with representations that steer cleanly but are not clearly shown here to carry the hiring decision. The later checkpoints complicate that picture: all three Gemma-4 models have a race–competence interval that excludes zero, Gemma-4-26B-A4B additionally has a gender–warmth interval that excludes zero, and the two Qwen3.6 checkpoints show a comparable spread across warmth and competence, matching the pattern already present in Qwen3-14B. These unadjusted patterns suggest that the absence of mediation may be specific to the Gemma-3 generation, but they do not establish that model-generation difference confirmatorily.

Discussion and Conclusion

One extraction procedure, applied without modification to nine checkpoints from three model families, recovers warmth and competence directions in every one of them. The probes reach ceiling accuracy on held-out topics, their effect sizes stand far outside a random-direction

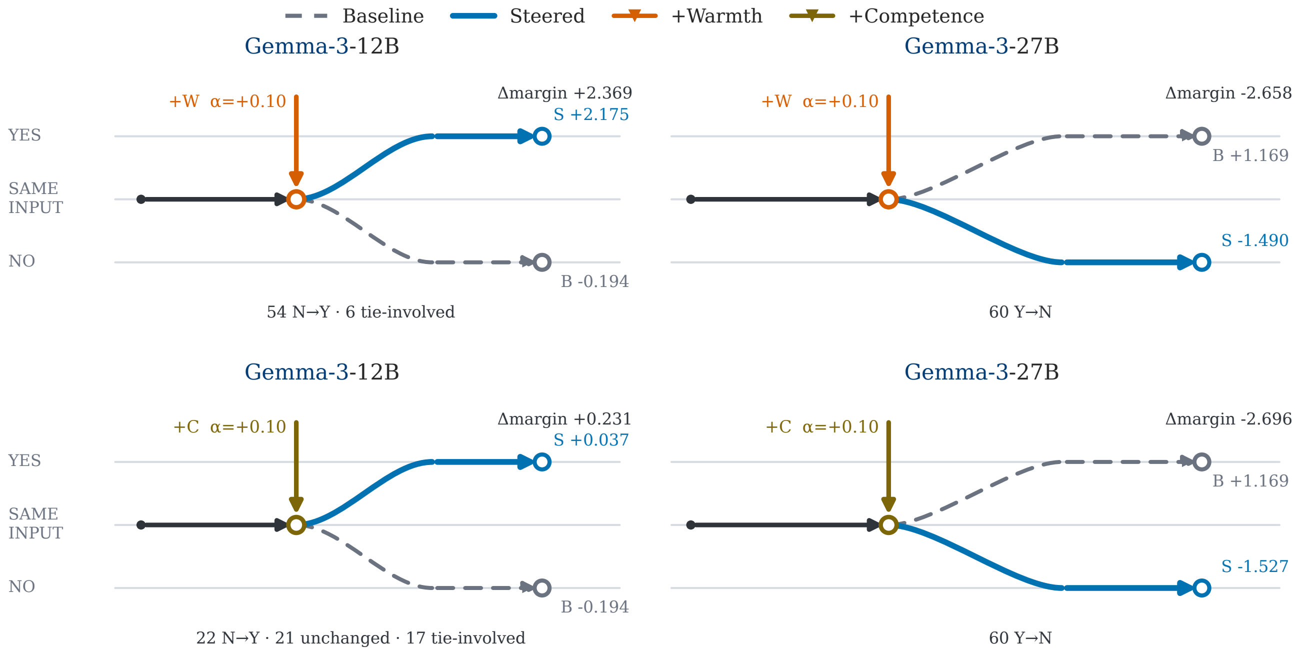

Figure 7: Matched examples show that positive concept steering can move callback decisions in either direction. Rows show warmth and competence; columns compare Gemma-3-12B, whose mean decision moves from No to Yes, with Gemma-3-27B, whose mean decision moves from Yes to No. All panels use the same 60 applications and $\alpha = +0.10$. Gray dashed paths show baseline outcomes, blue paths show steered outcomes, and bottom annotations summarize individual-name transitions. The panels were selected to display both transition directions under matched conditions, not to estimate their prevalence. Endpoint margins, strengths, and transition counts are empirical; connector geometry is schematic. Complete results for all nine models and both axes appear in Tab. S.10.

null, and rebuilding a direction from disjoint halves of the corpus returns nearly the same vector. Read on its own, that pattern suggests a stable and widely shared representation of social perception.

The rest of our results complicate that reading. What the recovered directions correspond to, and what happens when they are pushed, varies across checkpoints in ways the extraction statistics do not anticipate. Warmth- and competence-associated contrastive directions are recoverable in all nine models, but what those directions correspond to cannot be assumed identical from one model to the next.

Warmth Is the Less Stable Axis. Competence behaves as the more orderly of the two dimensions throughout. Its split-half reconstructions are more stable than warmth in every model without exception, and its correlation with human name ratings is positive and significant in seven of the nine, with the remaining two small, negative, and nonsignificant. Warmth does not follow suit. Four checkpoints return a warmth correlation that is negative or indistinguishable from zero, with Llama-3.1-8B at $\rho = -0.287$ while Gemma-3-27B reaches $+0.388$ on the same names and the same rating data.

A sign difference of that size is not easily read as noise. Both models separate our warmth stories cleanly, so the direction is doing something consistent inside each model; the two simply disagree about which names it should favor. One reading is that warmth, which the Stereotype Content Model ties to perceived intent rather than to observable outcomes, has weaker and more contested textual regularities than competence, so its encoding depends more heavily on the particular training corpus and alignment procedure. We cannot test that account with these data, and offer it as a hypothesis rather than a conclusion.

Naming a Direction Is a Claim We Make. That a vector separates warmth-contrastive stories does not by itself establish that it encodes warmth as people understand the term. The label comes from the stimuli we wrote, not from the model. In four checkpoints the direction so labeled does not track how people actually rate the warmth of a name, which suggests the construct governing model behavior on our stimuli may not be the construct the label names.

This distinction seems worth taking seriously for auditing practice. A behavioral audit measures a quantity whose meaning is fixed outside the model, such as whether names associated with one group receive fewer callbacks. An audit conducted at the level of internal representations measures something whose interpretation has to be argued for, and our results suggest that argument may need to be remade for each model rather than established once for a method. Proposals to assess systems by inspecting their internal social representations would therefore seem to carry a validation burden that outcome-based auditing does not.

Response to Intervention Is Range-Dependent. Fitting a line through each model-axis steering response describes the data well in only seven of eighteen cases. The departures are not uniform. Gemma-3-27B holds R^2 below 0.35 on both axes and ends at a negative delta despite a positive fitted slope, and comparable sign changes between moderate and larger interventions appear in several later checkpoints. The Qwen3-14B competence response makes the range issue especially compact: its broad-grid slope is slightly negative, while the local -0.10 to $+0.10$ change is positive and its $+0.10$ endpoint effect is only $+0.010$. The baseline in these cases leaves room to move, so saturation alone does not account for the pattern. A direction that interacts with other representations, rather than acting on an isolated axis, would produce this behavior more readily than one that scales cleanly.

The practical consequence is that a claim of the form "steering this direction shifts the callback margin upward" may hold only within the range where it was measured. An intervention whose sign depends on its magnitude is difficult to treat as a mitigation without evidence that it remains stable across the range in which it would operate. Reporting an effect at one chosen strength can conceal the property that determines whether the intervention behaves predictably.

Steerability and Mediation Come Apart. External steerability and internal mediation do not track each other in our results. Gemma-3-12B carries the largest normalized warmth steerability yet has no mediation interval that excludes zero at the unadjusted 95% level. Llama-3.1-8B, whose concept-level response is far smaller, has a race–warmth indirect effect of $+0.190$ with a 95% interval of $[+0.111, +0.292]$. Responding strongly to an imposed intervention and routing a name-group signal through that same representation unprompted may therefore be different properties. The two Gemma-3 checkpoints have no unadjusted interval excluding zero on either grouping, whereas the Gemma-4 and Qwen3.6 checkpoints have several. Because no multiple-comparison adjustment is reported, this contrast is exploratory and does not identify a confirmed model-family difference.

Limitations. This study has several limitations, grouped below by the aspect of the design each one bears on.

Stimuli and probe construction. The directions are only as good as the material and the layer they are built from, and both involve choices that were fixed in advance rather than optimized.

- **Circularity in stimulus generation**: The concept story corpus was generated by a large language model, raising a concern about circularity in a study that probes language model representations. The generating model (Claude, Anthropic) and the probed models (Gemma, Qwen, Llama) come from different organizations and architectures, so the warmth/competence contrasts are not trivially reproduced by a shared generator. Even so, AI-generated stimuli could share stylistic properties that inflate probe accuracy across model families; replication with human-authored or template-based stimuli is desirable.
- **Limited topic coverage**: The corpus covers 50 topics. Broader topic coverage and larger per-condition samples would further test how far the directions generalize.
- **The corpus cannot be regenerated exactly**: Stories were written interactively rather than by a batch script, and the prompt was refined during generation, most consequentially by adding the identity and off-axis constraints after an audit of an early pilot. The prompts as issued are reported in Tab. S.2, but no seed or transcript fixes the sampling, so re-running them would produce a different corpus. The 200 stories are released in full and every downstream result derives from that fixed text, so the analyses reported here are reproducible even though the stimulus generation is not.
- **A fixed probe-layer heuristic, not validated per model**: We set probe_layer_frac = 0.66 for every model based on the relative depth reported for emotion representations in Sofroniew et al. [23], applying it as a static, prespecified choice rather than testing separability at that depth on each model individually. A full-depth layer sweep on the training story corpus shows that this heuristic layer rarely coincides with the layer of peak separability: at Gemma-3-12B the selected layer reaches a warmth Cohen's d of 2.68, while the best-separated layer in the same sweep reaches 6.27; Gemma-3-27B (2.95 versus 5.10) and Gemma-4-31B (7.56 versus 11.49) show comparable gaps. This mismatch does not fully explain the weak or inverted name-level correlations reported above: Gemma-4-12B already reaches a high separability score at its selected layer ($d = 8.63$), yet the resulting warmth probe fails to correlate with human name ratings ($\rho = +0.009$, n.s.). Story-level separability, which guided layer selection, therefore does not

guarantee that the resulting direction generalizes to bare applicant names, a case the training stories never present. Validating or tuning the probe layer per model, rather than transferring a single depth from a different study, is likely more principled; whether averaging or weighting probe projections across several separable layers would recover a stronger name-level signal than the single fixed layer is a question we leave open.

- **Ceiling effects in cross-validated accuracy**: 5-fold and topic-holdout accuracy reach 1.000 with zero variance across folds for every model and both axes. This is consistent with the large Cohen's d values reported above (for example $d = 2.68$ for warmth at Gemma-3-12B), but a ceiling this uniform is also expected on statistical grounds alone: with only 100 stories per axis and residual-stream vectors of several thousand dimensions, near-perfect linear separability is likely regardless of whether the direction captures a meaningful concept. Topic-holdout accuracy was designed specifically to close the topic-vocabulary leakage route available to plain 5-fold splitting, described above, yet it saturates at the same ceiling; each topic-holdout fold still contains only around ten held-out topics against several-thousand-dimensional vectors, so the same high-dimension, low-sample-size argument applies there as well, and the stricter split does not by itself rule it out. Cross-validated accuracy at ceiling therefore cannot distinguish between models or axes on its own, and should be read alongside the effect-size and split-half checks rather than as independent evidence of signal strength.

What the directions measure. Separating the stories a direction was built from does not establish that the direction encodes the construct its label names, and several results bear directly on that gap.

- **Moderate probe-to-human alignment**: Across all nine models (Tab. 5), warmth correlations with human ratings range from $\rho = -0.287$ to $+0.388$ and competence correlations range from $\rho = -0.058$ (n.s.) to $+0.466$. Competence tracks human perception more consistently than warmth, which turns negative or non-significant in four of the nine models; the models with the strongest warmth alignment (Gemma-3-12B, Gemma-3-27B) reach only moderate correlations even at their best. Taken together, the model's internal probes capture a component of human social perception rather than replicating it faithfully, and warmth is the less reliably captured of the two constructs. The subset with significant negative warmth alignment is discussed separately below.
- **Significant negative name-level warmth alignment in three models**: Warmth scores correlate negatively with human ratings in Llama-3.1-8B ($\rho = -0.287$), Gemma-4-26B-A4B ($\rho = -0.190$), and Qwen3-14B ($\rho = -0.178$), with all three correlations significant. Results involving name-level warmth for these models should be interpreted with this negative correspondence in mind. This evidence concerns the relation between the extracted score and human ratings; it does not by itself establish that the model encodes a fully inverted warmth construct.
- **Direction specificity is not established**: In the six-direction comparison the mean-difference direction is the largest intervention in only one of four model-axis rows, and in the three Gemma-4 checkpoints the calibrated random control's interval contains the dense-target effect. Steering the labeled direction moves the judgment, but these controls do not show that the effect belongs to that direction rather than to a comparable perturbation in the same region. The control basis is also uneven: five checkpoints carry 99 SD-matched random directions while the other four carry a single random direction, so the comparison is not equally strong across models.

Intervention and measurement precision. The causal results depend on an intervention whose calibration and an outcome whose resolution both have known limits.

- **Fragile causal effect at scale**: The causal warmth steering effect at the hiring level is positive and strong at 12B but non-monotone and fragile at 27B, so the causal claim does not generalize across scale without qualification.
- **Steering-strength range calibrated on two models**: We fixed the local steering range after testing saturation on Gemma-3-12B and Gemma-3-27B only, then applied it uniformly to all nine models; a more thorough methodology would calibrate this range separately for each model rather than reusing one pair's calibration across architectures that differ substantially in scale.
- **Callback-margin resolution**: Callback margins have finite, model-dependent resolution under reduced-precision inference. Several checkpoints, including both Gemma-3 models, Qwen3-14B, and both Qwen3.6 models, fall on a 0.125-unit grid in the canonical 282-name audits, while Llama-3.1-8B and the Gemma-4 models exhibit different or finer spacing. Casting the Yes/No logits to float32 before subtracting them prevents an additional rounding step, but it cannot recover precision already lost when the logits were produced. Full float32 inference would roughly double GPU memory usage and is likely infeasible for the 27B checkpoint on the available cluster hardware, so it was not attempted here.
- **Narrow callback variance at some checkpoints**: At Gemma-3-12B the callback margin standard deviation is 0.15 with only eight unique values, mak-

ing group-level disparity estimates at 12B unreliable; the 12B R4 results should not be cited as empirical findings. Llama-3.1-8B shows a comparably narrow spread (SD = 0.12, 12 unique values), so its group-level gaps should be read with the same caution. At Gemma-3-27B the larger variance (SD = 0.43, 20 unique values) makes group gaps of approximately 0.5 SD detectable, but the grid structure introduces discrete steps in all reported gap magnitudes. Qwen3-14B (SD = 0.35, 17 unique values) falls in the same usable range as 27B.

- **Open architectural question**: The architectural source of the elevated warmth/competence vector cosine observed in Gemma-3 relative to Qwen3 and Llama-3.1 across all network depths remains an open question.

Scope of inference. What these results can be generalized to is bounded by the testing regime, the benchmark, and the vectors the name-level analyses were run on.

- **Uncorrected mediation tests**: Across all nine models, the thirty-six mediation tests are presented uncorrected for multiple comparisons. Fourteen unadjusted 95% bootstrap intervals exclude zero, but no multiplicity-adjusted inference is reported and no corrected winner is identified. These paths should therefore be treated as exploratory.
- **Simplified hiring prompt**: The hiring prompt is simplified relative to real-world screening contexts (single role, fixed qualifications), and results may differ under richer evaluation settings or different job types.
- **Raw vectors only in the disparity and mediation pipeline**: The probe-vs-human correlations, group disparities, and mediation tests all use the raw dense directions; no denoised counterpart is computed for this pipeline, unlike the concept-level and hiring-steering causal tests, both of which report a denoised robustness check. Whether the disparity and mediation results are robust to the same PCA-denoising procedure is left open.
- **Benchmark and construct scope**: Both the callback rates and the name ratings come from North American sources, and the warmth and competence axes derive from a research tradition built largely on North American samples. Alignment measured here is alignment with the judgments that dataset records, which is narrower than alignment with human social perception in general.

Future Work. The immediate next step is expanding the hiring stimuli to additional job roles to test whether results generalise beyond administrative assistant. The SAE decomposition performed here on Gemma-3 can be extended to Llama-3.1 using the Llama Scope sparse autoencoders [15], enabling a model-family comparison at the feature level. Validating the concept probes against a human-authored warmth/competence dataset would address the circularity concern associated with AI-generated stimuli. On the social-science side, extending the hiring stimuli beyond names to category-level signals (age, disability, religion) would broaden the scope of the benchmark comparison and test whether the representational account generalises beyond race and gender. The crossed race-by-gender breakdown reported under Crossed Race × Gender Disparity, All Nine Models in the Supplementary Materials suggests the gender gap is not independent of race in most checkpoints, and that the two effects point in opposite directions across model families. Establishing whether that interaction is real would require standardized effects and interval estimates on larger cells than the present name panel provides. Translating these findings into practical debiasing interventions and evaluating their robustness to adversarial prompting is a natural next step for applied work.

Conclusion. Warmth and competence can be recovered as linear directions from nine open-weight models using one unmodified procedure, and pushing those directions changes the judgments the models state. Beyond that shared foundation the models diverge. They disagree on whether the warmth direction points the way human raters do, on whether a stronger intervention amplifies or reverses a weaker one, and on whether the representation carries a demographic signal into a hiring recommendation at all. A direction that separates our warmth and competence stories exists in all of these systems; what it corresponds to appears to be a property of the individual model. Findings obtained from a single checkpoint, our own included, should be treated as claims about that checkpoint until they are shown to replicate.

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Supplementary Materials

Details on Methods

Compute and Hardware

All experiments ran on two compute environments. The SCCKN cluster (Universität Konstanz) supplied NVIDIA RTX PRO 6000 Blackwell GPUs (96 GB VRAM), which ran the Gemma-3, Llama-3.1, and Qwen3 jobs (Gemma-3-12B, Gemma-3-27B, Llama-3.1-8B, Qwen3-14B) and most of the extraction and steering jobs for the Gemma-4 and Qwen3.6 checkpoints. Gemma-4-12B is a partial exception: its Stage 1 activations were first extracted on an NVIDIA L40, and a later retry ran on an L40S instead. A dedicated reproducibility audit on the exact L40 hardware class found that both device classes agree on which layer peaks for warmth and competence (layers 26 and 27, respectively), but differ non-negligibly at some layers (maximum competence Cohen's d difference of 0.914, cosine difference of 0.095). The exact-L40 run, hardware-matched to the original Stage 1 extraction, is the one reported throughout the main text. Select calibrated-steering replications for the Gemma-4 and Qwen3.6 checkpoints (all three Gemma-4 models and both Qwen3.6 models) instead ran on NVIDIA H100 GPUs at the Collective Computational Unit (CCU).

Story Corpus by Topic

Table S.1: Story corpus by topic ($n = 50$). Every topic contributes exactly one story to each of the four conditions (high warmth, low warmth, high competence, low competence), for 200 stories total, and every protagonist is nameless and referred to only as "they" throughout (see Story Preparation). The word-count range gives the shortest and longest of a topic's four stories. The Category column groups topics into the ten domains named in Story Preparation; these labels are our own assignment for readability and are not a field in the underlying stimuli data.

No.	Topic	Category	Words
1	A team meeting where a decision needs to be made under time pressure	Workplace	130–142
2	Receiving critical feedback from a supervisor in front of colleagues	Workplace	100–104
3	Presenting results that did not go as expected to senior management	Workplace	95–103
4	Discovering that a colleague made a serious mistake before the deadline	Workplace	99–103
5	Mentoring a junior colleague on their first week at the job	Workplace	97–105
6	Handling an escalating complaint from a dissatisfied customer	Workplace	93–100
7	Negotiating project scope with a client who keeps changing requirements	Workplace	91–92
8	Organising a company event with a limited budget and unclear instructions	Workplace	89–96
9	Taking over a project mid-way when the previous lead leaves unexpectedly	Workplace	90–101
10	Responding to an emergency system failure at 2am on call	Workplace	92–102
11	Being asked to do something you believe is wrong by a manager	Learning	91–102
12	Learning a completely new technical skill under pressure and alone	Learning	97–104
13	Debugging a complex problem with no documentation and no internet	Learning	94–99
14	Preparing for a high-stakes exam with only two days left	Learning	93–97
15	Trying to understand a subject that feels overwhelming at first	Learning	93–101
16	Explaining a technical concept to someone with no background in it	Learning	126–144
17	Helping a stranger who dropped their groceries in the street	Social	88–97
18	A difficult conversation with a family member about money	Social	90–100
19	Being the only person at a party who does not know anyone	Social	92–95
20	Comforting a friend who just received bad news	Social	89–102
21	Waiting for important medical test results	Health	92–109
22	Supporting a parent who is beginning to lose their independence	Health	90–95
23	Managing a chronic illness while trying to maintain a normal schedule	Health	88–96
24	Recovering from an injury when you were counting on being active	Health	90–105
25	Organising a neighbourhood clean-up with volunteers who cancel last minute	Community	129–140
26	Attending a community meeting where people strongly disagree	Community	91–100
27	Volunteering at a food bank on an unexpectedly chaotic shift	Community	89–99
28	Running for a small local position against an experienced opponent	Community	93–106
29	Advocating for a change at the local school as a parent	Community	90–101

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No.	Topic	Category	Words
30	Deciding whether to take a pay cut for a job you believe in	Finance	88–95
31	Realising mid-month that you have far less money than expected	Finance	90–104
32	Negotiating a salary for the first time in a new job offer	Finance	90–103
33	Dealing with a billing dispute that has lasted several months	Finance	89–103
34	Coaching a sports team through a long losing streak	Sport & Creative	129–143
35	Performing music in public for the first time and making a mistake	Sport & Creative	90–101
36	Finishing a creative project that has been stalled for months	Sport & Creative	97–107
37	Being a referee in a game where the players are friends	Sport & Creative	93–98
38	Entering a competition knowing you are not the favourite	Sport & Creative	91–107
39	Missing a connecting flight with no phone battery and no local currency	Travel	125–142
40	Navigating a city you have never been to, in a country whose language you do not speak	Travel	90–100
41	Being stranded overnight due to a transport strike with strangers	Travel	93–107
42	Starting over in a new country with no local contacts	Travel	91–99
43	Arriving at accommodation that is nothing like what was advertised	Travel	93–102
44	Realising your important data was lost and there is no backup	Technology	90–95
45	Trying to set up a complex technical system with no instructions	Technology	95–105
46	Being asked to fix someone else's code under a tight deadline	Technology	91–100
47	Teaching an elderly relative how to use a new device they are afraid of	Technology	90–102
48	Apologising sincerely to someone you have genuinely hurt	Personal Milestones	94–104
49	Being the first person in your family to pursue a university degree	Personal Milestones	92–102
50	Caring for a newborn alone for the first time with no experience	Personal Milestones	90–98

Story Generation and System Prompt

The corpus was written manually, condition by condition, by Claude Opus 4.8 against a fixed specification: a single nameless protagonist referred to only as "they," a per-condition list of forbidden words that would name the concept directly, a show-don't-tell instruction to convey warmth or competence through actions rather than labels, and the 90–150 word length band enforced across every condition. Tab. S.2 reserves the exact generation and system prompt text.

System prompt
You are a creative writer producing short vignettes for a psychology study. Follow the instructions precisely. Write only the paragraph — no titles, no explanations, no labels.
Per-story prompt
Write a single paragraph (90–150 words) about the following situation. Situation: {topic} The paragraph should describe ONE person's behaviour in this situation. That person should clearly exhibit {condition_description}. Critical rules: 1. Show the quality through the person's specific actions, decisions, and reactions — NEVER state it directly. 2. Do NOT use any of these words (or close variations): {forbidden_words}. 3. The paragraph should feel natural, like a scene from a realistic short story. 4. Do not name the quality at any point. 5. The protagonist must be nameless and referred to only as "they," with no cues to gender, ethnicity, or national origin. 6. Hold the other dimension at an ordinary level: when writing warmth, the person should read as neither notably competent nor notably incompetent, and the reverse when writing competence. 7. Write only the paragraph. No title, no label, no explanation.
{condition_description}
high warmth: strong warmth — genuine care, attentiveness, and concern for others *low warmth*: low warmth — emotional distance, self-focus, and indifference to others *high competence*: high competence — effectiveness, careful thinking, and reliable execution *low competence*: low competence — disorganisation, poor judgement, and repeated errors
{forbidden_words}
high warmth: warm, warmth, kind, kindness, caring, empathetic, empathy, compassionate, compassion, friendly, friendliness *low warmth*: cold, coldness, unfriendly, distant, aloof, uncaring, indifferent, hostile *high competence*: competent, competence, skilled, capable, capability, expert, expertise, proficient, proficiency, efficient, efficiency *low competence*: incompetent, incompetence, unskilled, incapable, inefficient, inefficiency, clumsy

Table S.2: Prompts used to generate the concept-story corpus. The corpus was produced interactively with Claude Opus 4.8 rather than by a batch script, and rules 5 and 6 were added during generation after an audit of an early pilot found that unconstrained prompting assigned male, Anglo-named protagonists to the low-warmth and low-competence conditions. The repository's `src/generate_stimuli.py` holds a batch template for a larger corpus that was specified but not run; it differs from the text above in its requested length band and does not carry rules 5 and 6. Realized story lengths were 88 to 144 words, mean 100.

Example Concept Stories

Tab. S.3 quotes one story from each of the four conditions, all four sharing a single topic, so that the contrast between conditions is visible independent of the situation described.

Table S.3: Example stories, one per condition, sharing a single topic (topic 1 of Tab. S.1). The complete set of all 200 story texts is available in the project repository at `data/stimuli/concept_stories.jsonl`.

Condition	Story (topic: "A team meeting where a decision needs to be made under time pressure")
High warmth	Noticing that one teammate hadn't spoken since the start, they paused and asked what that person was seeing from their side of the project. When two colleagues talked over each other, they waited, then restated each point so neither felt brushed aside. Aware that everyone had skipped lunch, they pushed the open box of crackers to the middle of the table. As the deadline pressed in, they made sure the quietest intern's worry about the rollout went up on the board alongside everyone else's. 'We decide together or we don't decide well,' they said, watching faces rather than the clock. Once the group chose a direction, they stayed behind to walk the teammate who'd looked lost through what had been agreed, and afterwards messaged the intern to say the question raised had changed the final plan.
Low warmth	Opening the meeting, they announced they already knew the right call and that the discussion was a formality. When a colleague tried to raise a risk with the vendor, they kept scrolling their phone and said there was no time for hypotheticals. An intern asked a clarifying question; they answered someone else, as though it hadn't been spoken. They read the group's hesitation as wasted time rather than doubt and pushed for a vote. 'I don't need everyone comfortable, I need a decision,' they said, not looking up. When a teammate mentioned being away during the rollout and feeling worried, they replied that it wasn't their problem to manage. The group agreed quickly, mostly to end the meeting, and they left first without waiting for anyone to gather their things.

continued on next page

Condition	Story (topic: "A team meeting where a decision needs to be made under time pressure")
High competence	Having read the vendor report the night before, they broke the stall by putting three options on the board, each already filled in with its cost and failure risk. They flagged that the cheapest option hid a renewal clause that would double the price in year two, buried on page forty. When someone proposed a fourth path, they tested it aloud against the deadline and showed exactly where it broke. They set a ten-minute limit per section and tracked it, steering the group back whenever it drifted. Before the vote they summarised the trade-offs in two sentences so no one chose blind. Once the decision was made, they assigned every follow-up to a named owner with a date, typed it into the shared doc while the group watched, and read it back. The meeting finished four minutes early, every action already logged.
Low competence	They started the meeting before locating the file, then spent minutes searching their inbox while the room waited. The numbers they finally opened were from the previous quarter, though they insisted these were current until someone pointed at the date. They proposed the option with the highest upside without mentioning its cost, and when asked about the budget guessed a figure that proved wrong by half. Each risk that came up went onto a sticky note they then misplaced. They changed the agenda twice mid-discussion, reopening debates already settled. With ten minutes left they realised the one person who owned the rollout had never been invited. The group agreed to revisit it the next day, having decided nothing, and they promised a summary that by morning contradicted what most people remembered agreeing.

Probe Validation Checks

Each check below states what it tests and the reasoning behind it. Results for all nine models are summarized in Methods.

- **5-fold cross-validated accuracy**: We need to know whether the direction we found is picking up on something real, or whether it just memorized the exact 200 stories we happened to write. The 100 warmth stories (50 high, 50 low) are split into 5 equal groups of 20. In each round, a classifier is trained on 4 of the groups and tested on the 5th group, which it has never seen; this repeats 5 times so every group is tested once. Guessing randomly would get the label right about 50% of the time, so a classifier that gets far more than half of these unseen stories right is reading a real high-versus-low pattern in the stories, not reciting stories it has already seen.
- **Topic-holdout accuracy**: The check above still lets a classifier study some stories about a topic, say a stressful team meeting, before being tested on other stories about that same topic; getting those right could mean it recognized the topic, not the tone. This check closes that gap. The 50 topics are grouped so that every story about a topic, high- and low-condition alike, is held out together, and each test round contains topics the classifier has never read a single sentence about, in either condition. A classifier that still gets these completely unfamiliar topics right is recognizing warmth or competence itself, not the leftover vocabulary of a topic it has already studied.
- **Cohen's d**: The two checks above only tell us whether high- and low-condition stories can be told apart, not how far apart they really are. Each story is projected onto the direction, reducing it to a single number, and Cohen's d measures the gap between the high- and low-condition averages on that number line. To know whether that gap is meaningful, we generate 1,000 random directions instead of the real one and compute the same gap for each. Every one of those 1,000 random directions produces a much smaller gap than our real direction does, which means the separation is not a coincidence.
- **Split-half cosine stability**: A direction built from one arbitrary set of stories should not depend on exactly which stories happened to be written. Each condition's 50 stories are randomly split into two non-overlapping halves of 25, and a separate direction is built from each half using only that half's stories. The cosine between these two independently-built directions tells us how similar they are. Across all nine models, this cosine is consistently high (roughly 0.69 to 0.82 for warmth, 0.83 to 0.90 for competence), meaning the direction reflects a stable property of the concept rather than an artifact of which 25 stories happened to end up in each half.
- **Cross-axis classification**: If warmth and competence were fully independent, knowing a story's position along the warmth direction would tell us nothing about whether it is high- or low-competence. To test this, competence stories are projected onto the warmth direction, and that single number is used to guess the competence label; the same test is then run in the other direction. Across all nine models this guess is highly accurate, typically 0.82 to 1.00, which means the two directions are not fully independent: they share a substantial amount of evaluative content rather than capturing two fully separate constructs.

PCA Denoising, All Nine Models

The warmth and competence direction vectors share a substantial common component that reflects general narrative valence: stories that portray a protagonist positively (whether warmly or competently) occupy a similar region of the residual stream, and the same holds for negatively valenced stories. To separate conceptual content from this shared evaluative polarity, we follow the neutral-corpus denoising procedure of Sofroniew et al. [23]. We build a neutral corpus of 1,500 length-matched introductory passages drawn from distinct Wikipedia articles (90–200 words) and filtered to exclude emotionally charged or violent content, then extract residual-stream activations at the same layer and pooling rule used for the concept stories. We then fit principal component analysis (PCA) on these neutral activations and project out, from both direction vectors, the top components that together explain at least 50 percent of neutral variance. For Gemma-3-12B, the first principal component alone explains 56.1 percent of neutral variance, and removing it lowers the cosine between the warmth and competence vectors from 0.749 to 0.530. For Gemma-3-27B, 43 components are needed to cross the 50 percent threshold (50.2 percent explained), lowering the inter-axis cosine from 0.708 to 0.487. The remaining post-denoising cosine, 0.53 and 0.49 respectively, is close to the genuine warmth-competence correlation observed in human ratings ($r \approx 0.61$; [9]), which suggests that the residual overlap reflects real conceptual co-variation rather than a failure of the denoising step. The denoised vectors serve as a robustness check; the raw dense directions define the primary cross-model causal comparisons and the callback-transition census reported in the main text. Tab. S.4 extends this same procedure to all nine models.

Model	k (PCs removed)	Variance kept	cos(W,C) before	cos(W,C) after
Gemma-3-12B	1	56.1%	0.749	0.530
Gemma-3-27B	43	50.2%	0.708	0.487
Llama-3.1-8B	48	50.3%	0.505	0.492
Gemma-4-12B	11	51.2%	0.494	0.473
Gemma-4-26B-A4B	11	50.3%	0.587	0.564
Gemma-4-31B	17	50.6%	0.526	0.469
Qwen3-14B	19	50.5%	0.536	0.510
Qwen3.6-27B	27	50.2%	0.580	0.560
Qwen3.6-35B-A3B	17	50.3%	0.619	0.595

Table S.4: Neutral-corpus PCA denoising, all nine models. k is the number of leading principal components of 1,500 neutral Wikipedia activations removed from both direction vectors, the smallest k crossing the 50% neutral-variance threshold. cos(W,C) before/after is the cosine between the warmth and competence directions, measured before and after removing those components. Denoising reduces cos(W,C) in every model but never eliminates it.

Cross-Model Story-Ranking Agreement, All 36 Pairs

Tab. S.5 lists the Spearman correlation, both overall and within-condition, for every one of the $\binom{9}{2} = 36$ possible model pairs on both axes, the full table behind Fig. 5 in Results.

Table S.5: Cross-model Spearman story-ranking agreement, all 36 model pairs. For each pair and axis, every one of the 200 concept stories is projected onto both models' own warmth or competence direction, and the two resulting score lists are correlated with Spearman's ρ. "Overall" includes the high/low condition separation itself; "within-condition" restricts the correlation to stories sharing one condition, so it reflects agreement on finer-grained story ordering rather than the coarse high-versus-low split. The four per-condition ρ values behind the within-condition summary are in `results/tables/cross_model_agreement_9model.csv`.

Model A	Model B	Overall ρ	Within-condition ρ
Warmth			
Gemma-3-12B	Gemma-3-27B	0.818	0.555
Gemma-3-12B	Llama-3.1-8B	0.768	0.203
Gemma-3-12B	Gemma-4-12B	0.766	0.195
Gemma-3-12B	Gemma-4-26B-A4B	0.741	0.095
Gemma-3-12B	Gemma-4-31B	0.801	0.285
Gemma-3-12B	Qwen3-14B	0.760	0.160
Gemma-3-12B	Qwen3.6-27B	0.803	0.308
Gemma-3-12B	Qwen3.6-35B-A3B	0.797	0.251
Gemma-3-27B	Llama-3.1-8B	0.783	0.208
Gemma-3-27B	Gemma-4-12B	0.757	0.120
Gemma-3-27B	Gemma-4-26B-A4B	0.759	0.124
Gemma-3-27B	Gemma-4-31B	0.762	0.182

Tab. S.5 continued

Model A	Model B	Overall ρ	Within-condition ρ
Gemma-3-27B	Qwen3-14B	0.777	0.223
Gemma-3-27B	Qwen3.6-27B	0.792	0.308
Gemma-3-27B	Qwen3.6-35B-A3B	0.760	0.174
Llama-3.1-8B	Qwen3-14B	0.978	0.833
Llama-3.1-8B	Qwen3.6-27B	0.945	0.641
Llama-3.1-8B	Qwen3.6-35B-A3B	0.896	0.541
Gemma-4-12B	Llama-3.1-8B	0.931	0.476
Gemma-4-12B	Gemma-4-26B-A4B	0.911	0.434
Gemma-4-12B	Gemma-4-31B	0.940	0.574
Gemma-4-12B	Qwen3-14B	0.941	0.516
Gemma-4-12B	Qwen3.6-27B	0.920	0.453
Gemma-4-12B	Qwen3.6-35B-A3B	0.884	0.451
Gemma-4-26B-A4B	Llama-3.1-8B	0.932	0.586
Gemma-4-26B-A4B	Gemma-4-31B	0.905	0.467
Gemma-4-26B-A4B	Qwen3-14B	0.948	0.698
Gemma-4-26B-A4B	Qwen3.6-27B	0.919	0.530
Gemma-4-26B-A4B	Qwen3.6-35B-A3B	0.892	0.488
Gemma-4-31B	Llama-3.1-8B	0.908	0.428
Gemma-4-31B	Qwen3-14B	0.922	0.497
Gemma-4-31B	Qwen3.6-27B	0.926	0.502
Gemma-4-31B	Qwen3.6-35B-A3B	0.902	0.479
Qwen3-14B	Qwen3.6-27B	0.953	0.724
Qwen3-14B	Qwen3.6-35B-A3B	0.904	0.595
Qwen3.6-27B	Qwen3.6-35B-A3B	0.930	0.685
Competence			
Gemma-3-12B	Gemma-3-27B	0.825	0.557
Gemma-3-12B	Llama-3.1-8B	0.782	0.295
Gemma-3-12B	Gemma-4-12B	0.786	0.326
Gemma-3-12B	Gemma-4-26B-A4B	0.792	0.307
Gemma-3-12B	Gemma-4-31B	0.812	0.413
Gemma-3-12B	Qwen3-14B	0.795	0.364
Gemma-3-12B	Qwen3.6-27B	0.799	0.330
Gemma-3-12B	Qwen3.6-35B-A3B	0.785	0.282
Gemma-3-27B	Llama-3.1-8B	0.834	0.304
Gemma-3-27B	Gemma-4-12B	0.831	0.284
Gemma-3-27B	Gemma-4-26B-A4B	0.823	0.258
Gemma-3-27B	Gemma-4-31B	0.819	0.311
Gemma-3-27B	Qwen3-14B	0.836	0.314
Gemma-3-27B	Qwen3.6-27B	0.829	0.262
Gemma-3-27B	Qwen3.6-35B-A3B	0.800	0.204
Llama-3.1-8B	Qwen3-14B	0.992	0.899
Llama-3.1-8B	Qwen3.6-27B	0.979	0.757
Llama-3.1-8B	Qwen3.6-35B-A3B	0.943	0.513
Gemma-4-12B	Llama-3.1-8B	0.964	0.590
Gemma-4-12B	Gemma-4-26B-A4B	0.960	0.618
Gemma-4-12B	Gemma-4-31B	0.952	0.645
Gemma-4-12B	Qwen3-14B	0.970	0.646
Gemma-4-12B	Qwen3.6-27B	0.965	0.621
Gemma-4-12B	Qwen3.6-35B-A3B	0.935	0.476
Gemma-4-26B-A4B	Llama-3.1-8B	0.974	0.751
Gemma-4-26B-A4B	Gemma-4-31B	0.947	0.619
Gemma-4-26B-A4B	Qwen3-14B	0.980	0.802
Gemma-4-26B-A4B	Qwen3.6-27B	0.972	0.727
Gemma-4-26B-A4B	Qwen3.6-35B-A3B	0.945	0.532
Gemma-4-31B	Llama-3.1-8B	0.943	0.577
Gemma-4-31B	Qwen3-14B	0.952	0.663
Gemma-4-31B	Qwen3.6-27B	0.952	0.633
Gemma-4-31B	Qwen3.6-35B-A3B	0.928	0.515

Tab. S.5 continued

Model A	Model B	Overall ρ	Within-condition ρ
Qwen3-14B	Qwen3.6-27B	0.981	0.783
Qwen3-14B	Qwen3.6-35B-A3B	0.952	0.584
Qwen3.6-27B	Qwen3.6-35B-A3B	0.957	0.630

Gemma Scope Steering Robustness, Gemma-3-12B/27B

Three further checks probe the direction-specificity result reported in Results. Tab. S.6 first asks whether the Gemma Scope 2 decomposition itself is trustworthy: reconstruction cosine stays above 0.995 at every SAE width for both models, so the sparse representation preserves the original activation well, even though topic-holdout accuracy inside that sparse space and the decoded direction's alignment with the dense direction both vary more across widths.

Model	Width	Recon. cosine	Active feats.	Topic-CV (W/C)	Decoded cosine (W/C)
Gemma-3-12B	16k	0.997	8.6	0.94 / 0.78	0.73 / 0.72
Gemma-3-12B	65k	0.997	6.3	0.72 / 0.77	0.74 / 0.64
Gemma-3-12B	262k	0.998	5.6	0.70 / 0.69	0.75 / 0.72
Gemma-3-27B	16k	0.995	7.6	0.84 / 0.76	0.61 / 0.40
Gemma-3-27B	65k	0.995	6.9	0.77 / 0.75	0.58 / 0.57
Gemma-3-27B	262k	0.995	5.3	0.71 / 0.79	0.65 / 0.63

Table S.6: Gemma Scope 2 SAE reconstruction quality. Recon. cosine is the cosine between the original activation and its SAE reconstruction; active features is the mean number of nonzero SAE features per story; topic-CV is 5-fold topic-holdout accuracy for warmth/competence classification inside the sparse feature space; decoded cosine is the alignment between the decoded SAE direction and the raw dense direction.

Tab. S.7 removes, rather than adds, a feature group and tracks the resulting change in the high-low margin gap.

Model	Axis	Target axis	Shared	Opposing axis	Random
Gemma-3-12B	Warmth	+0.225	+0.287	+0.100	+0.013
Gemma-3-12B	Competence	-0.050	+0.113	+0.125	+0.000
Gemma-3-27B	Warmth	+0.075	-0.725	+0.044	+0.200
Gemma-3-27B	Competence	-0.087	-0.319	+0.013	+0.025

Table S.7: Feature ablation, Gemma-3-12B/27B. Change in the high-versus-low margin gap when the named group of SAE features is zeroed out rather than added to; a negative value means ablation shrinks the gap. Shared-feature ablation shrinks both gaps only in Gemma-3-27B and increases both gaps in Gemma-3-12B. The shared feature set therefore shows scale-specific necessity rather than a uniform mechanism across checkpoints.

Tab. S.8 asks whether the two Gemma-3 scales converge on similar sparse features in the first place, matching each Gemma-3-12B feature to its closest Gemma-3-27B counterpart and comparing their story-level response profiles against a permutation null.

Axis	n matches	Obs. mean	Obs. median	Null mean [95% CI]	p
Warmth	135	0.644	0.663	0.354 [0.324, 0.387]	0.0020
Competence	192	0.655	0.694	0.403 [0.365, 0.439]	0.0020

Table S.8: Cross-scale SAE feature matching, Gemma-3-12B $\leftrightarrow$ Gemma-3-27B. For each matched feature pair, story-profile correlation is compared against a 500-permutation null. Observed mean/median well above the null mean, with a small permutation p-value, indicates the two scales share features with correlated story-level responses beyond chance.

Model	Axis	$\alpha = -0.50$	-0.25	0	$+0.25$	$+0.50$
Gemma-3-12B	Warmth	-5.68	-3.30	+0.00	+7.29	+11.43
Gemma-3-12B	Competence	-4.09	+0.79	+0.00	+8.70	+7.32
Gemma-3-27B	Warmth	+0.03	-0.96	+0.00	+6.17	+14.44
Gemma-3-27B	Competence	+4.28	+1.23	+0.00	+3.68	+5.08

Table S.9: Concept-level steering at wide strengths. Change in the held-out Yes-versus-No logit margin under dense-target steering, Gemma-3-12B and Gemma-3-27B, the two models where the wide strength grid was tested. The step from +0.25 to +0.50 is smaller than the step before it for three of the four rows, consistent with the saturating breakdown Methods describes; Gemma-3-27B warmth is the exception, still accelerating at +0.50.

Table S.10: Complete positive-endpoint callback transitions. Mean baseline and steered callback margins are followed by their decision category in parentheses. Transition cells list every nonzero individual-name transition among No (N), Tie (T), and Yes (Y); each cell sums to 60.

			Warmth			Competence		
Model	α	Baseline	Endpoint	Δ margin	Transitions	Endpoint	Δ margin	Transitions
Gemma-3-12B	+0.10	-0.194 (N)	+2.175 (Y)	+2.369	54 N→Y; 6 T→Y	+0.037 (Y)	+0.231	19 N→N, 13 N→T, 22 N→Y 1 T→N, 2 T→T, 3 T→Y
Gemma-3-27B	+0.10	+1.169 (Y)	-1.490 (N)	-2.658	60 Y→N	-1.527 (N)	-2.696	60 Y→N
Llama-3.1-8B	+0.10	-2.064 (N)	-1.595 (N)	+0.469	60 N→N	-1.739 (N)	+0.325	60 N→N
Gemma-4-12B	+0.10	+17.188 (Y)	+18.237 (Y)	+1.049	60 Y→Y	+18.715 (Y)	+1.527	60 Y→Y
Gemma-4-26B-A4B	+0.10	+21.245 (Y)	+21.307 (Y)	+0.062	60 Y→Y	+20.865 (Y)	-0.380	60 Y→Y
Gemma-4-31B	+0.10	+25.615 (Y)	+25.173 (Y)	-0.442	60 Y→Y	+24.801 (Y)	-0.814	60 Y→Y
Qwen3-14B	+0.10	+1.673 (Y)	+2.096 (Y)	+0.423	60 Y→Y	+1.683 (Y)	+0.010	60 Y→Y
Qwen3.6-27B	+0.10	+5.346 (Y)	+6.542 (Y)	+1.196	60 Y→Y	+5.879 (Y)	+0.533	60 Y→Y
Qwen3.6-35B-A3B	+0.10	+3.031 (Y)	+3.998 (Y)	+0.967	60 Y→Y	+3.465 (Y)	+0.433	60 Y→Y

Additional Results

Wide-Strength Concept Steering, Gemma-3

The wide-strength calibration sweep is available for Gemma-3-12B and Gemma-3-27B only. It is retained here as the numeric basis for the saturation discussion in Results rather than presented inside the nine-model sequence.

Complete Positive-Steering Callback Transitions

Tab. S.10 reports every model-axis condition used in the positive-endpoint comparison. The table is the complete result; the matched panels in Fig. 7 are illustrative examples selected under a common family, application set, and steering strength. Gemma-3-12B, Gemma-3-27B, and the Qwen3.6 and Gemma-4 checkpoints use $\alpha = +0.10$, as do Llama-3.1-8B and Qwen3-14B. The common local endpoint makes all nine transition rows directly comparable in steering strength.

Positive steering crosses the mean decision boundary in both directions among the Gemma-3 models. Llama-3.1-8B remains in the No category for every name, while the other six models remain in the Yes category, even when the mean margin decreases. This pattern reflects two distinct mechanisms. First, a concept direction constructed from high-minus-low stories is not constrained to align positively with the callback readout. Second, models beginning far from zero can move substantially without changing the categorical recommendation. The Gemma-3-27B response adds a third complication because its local dose-response reverses between +0.05 and +0.10, indicating a narrow nonlinear intervention regime.

Marginal-Group Levels, All Nine Models

Tab. S.11 provides the levels underlying the main-text gap comparison. Human values are shared across models and therefore appear once; the model blocks retain the standardized representation scores and raw callback margins for every group.

Group	n	Human warmth/competence (z)	Human callback
Black	47	-0.30 / -0.58	0.183
White	180	+0.21 / +0.15	0.171
Female	154	+0.17 / -0.00	0.145
Male	115	-0.16 / +0.07	0.181

Model	Black	White	Female	Male
Gemma-3-12B	-0.44/-0.43	+0.45/+0.44	+0.11/+0.10	-0.04/-0.03
Gemma-3-27B	-0.39/-0.36	+0.35/+0.36	+0.11/+0.11	-0.08/-0.08
Llama-3.1-8B	+0.45/+0.24	-0.38/-0.22	+0.00/+0.18	-0.05/-0.23
Gemma-4-12B	-0.34/-0.40	+0.24/+0.44	-0.26/+0.01	+0.39/+0.07
Gemma-4-26B-A4B	+0.56/+0.36	-0.37/-0.39	+0.15/-0.13	-0.27/+0.02
Gemma-4-31B	-0.12/-0.19	+0.30/+0.36	-0.06/+0.04	+0.13/+0.02
Qwen3-14B	+0.35/-0.34	-0.46/+0.17	+0.13/+0.19	-0.28/-0.24
Qwen3.6-27B	+0.13/-0.24	-0.02/+0.15	-0.04/-0.29	+0.00/+0.34
Qwen3.6-35B-A3B	-0.15/+0.09	+0.31/+0.27	+0.24/+0.14	-0.24/-0.09

Model margin	Black	White	Female	Male
Gemma-3-12B	-0.213	-0.199	-0.165	-0.258
Gemma-3-27B	+1.665	+1.121	+1.118	+1.312
Llama-3.1-8B	-2.020	-2.067	-2.054	-2.107
Gemma-4-12B	+17.194	+17.172	+17.208	+17.175
Gemma-4-26B-A4B	+21.509	+21.095	+21.464	+20.902
Gemma-4-31B	+25.808	+25.577	+25.788	+25.408
Qwen3-14B	+1.758	+1.700	+1.851	+1.537
Qwen3.6-27B	+5.402	+5.328	+5.384	+5.284
Qwen3.6-35B-A3B	+2.981	+3.039	+3.143	+2.873

Table S.11: Underlying marginal-group levels. Human benchmark values are printed once because they are shared across models. The model grids report warmth/competence z-scores and raw unsteered callback margins for Black, White, female, and male name groups. The main-text gap comparison is in Tab. 7; study-specific raw values are in Tab. S.12.

Name-Level Warmth/Competence by Marginal Group, Raw Values by Study

Tab. S.12 is the raw-value companion to Tab. S.11 in the main text: the same race and gender marginal groups, broken down by source study rather than blended across it, since a name rated under more than one correspondence study can carry a substantially different callback rate across studies (Bertrand's "aisha" callback is 0.022, Kline's is 0.246). Model warmth and competence values are raw, unnormalized projections onto the concept direction, and human warmth and competence are the 0-100 Likert averages behind them, so neither is comparable in magnitude across models or against each other; the standardized (z-score) main-text version is built for that comparison.

Table S.12: Warmth/competence bias by marginal demographic group, broken down by source study, raw values. Companion to Tab. S.11: the same race and gender marginal groups, not blended across studies. Model warmth/competence are raw (unnormalized) projections and are not comparable in magnitude across models or against the 0-100 human warmth/competence scale; see Tab. S.11 for the standardized (z-score) comparison.

Model	Axis	Group × Study	n	Model warmth/competence	Human warmth/competence	Human callback
Gemma-3-12B	Race	Black / Bertrand	18	37751.85 / 40473.48	57.96 / 60.93	0.062
		Black / Kline	38	37704.47 / 40424.58	46.37 / 45.61	0.231
		White / Bertrand	18	38512.67 / 41287.53	55.30 / 61.95	0.094
		White / Kline	38	38533.82 / 41314.30	62.26 / 62.77	0.251
		White / Farber	12	38682.20 / 41473.86	58.41 / 60.51	0.111
		White / Neumark	122	38404.78 / 41176.39	53.69 / 55.70	0.168
		Female / Bertrand	18	38142.21 / 40888.71	60.75 / 62.87	0.083
		Female / Kline	38	38091.11 / 40834.26	55.49 / 52.14	0.241
		Female / Farber	12	38682.20 / 41473.86	58.41 / 60.51	0.111
		Female / Neumark	77	38474.98 / 41250.92	54.16 / 55.93	0.132
		Male / Bertrand	18	38122.30 / 40872.30	52.50 / 60.01	0.074
		Male / Kline	38	38147.17 / 40904.62	53.13 / 56.24	0.240
		Male / Neumark	45	38284.64 / 41048.87	52.88 / 55.30	0.229
Gemma-3-27B	Race	Black / Bertrand	18	26190.79 / 25237.29	57.96 / 60.93	0.062
		Black / Kline	38	26131.12 / 25178.45	46.37 / 45.61	0.231
		White / Bertrand	18	26754.42 / 25759.67	55.30 / 61.95	0.094
		White / Kline	38	26735.26 / 25740.82	62.26 / 62.77	0.251
		White / Farber	12	26786.33 / 25790.22	58.41 / 60.51	0.111
		White / Neumark	122	26556.63 / 25570.14	53.69 / 55.70	0.168
		Female / Bertrand	18	26476.82 / 25508.09	60.75 / 62.87	0.083
		Female / Kline	38	26407.84 / 25437.11	55.49 / 52.14	0.241
		Female / Farber	12	26786.33 / 25790.22	58.41 / 60.51	0.111
		Female / Neumark	77	26626.71 / 25638.55	54.16 / 55.93	0.132

Tab. S.12 continued

Model	Axis	Group × Study	n	Model warmth/competence	Human warmth/competence	Human callback
		Male / Bertrand	18	26468.39 / 25488.87	52.50 / 60.01	0.074
		Male / Kline	38	26458.54 / 25482.15	53.13 / 56.24	0.240
		Male / Neumark	45	26436.70 / 25453.09	52.88 / 55.30	0.229
Llama-3.1-8B	Race	Black / Bertrand	18	-1.27 / -0.16	57.96 / 60.93	0.062
		Black / Kline	38	-1.26 / -0.16	46.37 / 45.61	0.231
		White / Bertrand	18	-1.35 / -0.18	55.30 / 61.95	0.094
		White / Kline	38	-1.35 / -0.18	62.26 / 62.77	0.251
		White / Farber	12	-1.34 / -0.17	58.41 / 60.51	0.111
		White / Neumark	122	-1.34 / -0.17	53.69 / 55.70	0.168
		Female / Bertrand	18	-1.30 / -0.16	60.75 / 62.87	0.083
		Female / Kline	38	-1.30 / -0.16	55.49 / 52.14	0.241
		Female / Farber	12	-1.34 / -0.17	58.41 / 60.51	0.111
		Female / Neumark	77	-1.33 / -0.17	54.16 / 55.93	0.132
		Male / Bertrand	18	-1.31 / -0.18	52.50 / 60.01	0.074
		Male / Kline	38	-1.31 / -0.17	53.13 / 56.24	0.240
		Male / Neumark	45	-1.34 / -0.17	52.88 / 55.30	0.229
Gemma-4-12B	Race	Black / Bertrand	18	12.20 / 18.46	57.96 / 60.93	0.062
		Black / Kline	38	12.24 / 18.52	46.37 / 45.61	0.231
		White / Bertrand	18	12.31 / 19.06	55.30 / 61.95	0.094
		White / Kline	38	12.30 / 19.08	62.26 / 62.77	0.251
		White / Farber	12	12.26 / 19.06	58.41 / 60.51	0.111
		White / Neumark	122	12.29 / 18.98	53.69 / 55.70	0.168
		Female / Bertrand	18	12.25 / 18.73	60.75 / 62.87	0.083
		Female / Kline	38	12.25 / 18.74	55.49 / 52.14	0.241
		Female / Farber	12	12.26 / 19.06	58.41 / 60.51	0.111
		Female / Neumark	77	12.25 / 18.96	54.16 / 55.93	0.132
		Male / Bertrand	18	12.27 / 18.79	52.50 / 60.01	0.074
		Male / Kline	38	12.29 / 18.86	53.13 / 56.24	0.240
		Male / Neumark	45	12.36 / 19.01	52.88 / 55.30	0.229
Gemma-4-26B-A4B	Race	Black / Bertrand	18	1.06 / 5.53	57.96 / 60.93	0.062
		Black / Kline	38	1.07 / 5.56	46.37 / 45.61	0.231
		White / Bertrand	18	0.88 / 5.44	55.30 / 61.95	0.094
		White / Kline	38	0.87 / 5.43	62.26 / 62.77	0.251
		White / Farber	12	0.88 / 5.42	58.41 / 60.51	0.111
		White / Neumark	122	0.90 / 5.44	53.69 / 55.70	0.168
		Female / Bertrand	18	1.03 / 5.48	60.75 / 62.87	0.083
		Female / Kline	38	1.05 / 5.49	55.49 / 52.14	0.241
		Female / Farber	12	0.88 / 5.42	58.41 / 60.51	0.111
		Female / Neumark	77	0.92 / 5.44	54.16 / 55.93	0.132
		Male / Bertrand	18	0.91 / 5.48	52.50 / 60.01	0.074
		Male / Kline	38	0.89 / 5.49	53.13 / 56.24	0.240
		Male / Neumark	45	0.86 / 5.43	52.88 / 55.30	0.229
Gemma-4-31B	Race	Black / Bertrand	18	45.30 / 78.18	57.96 / 60.93	0.062
		Black / Kline	38	45.33 / 78.21	46.37 / 45.61	0.231
		White / Bertrand	18	45.59 / 79.01	55.30 / 61.95	0.094
		White / Kline	38	45.56 / 78.96	62.26 / 62.77	0.251
		White / Farber	12	45.48 / 78.87	58.41 / 60.51	0.111
		White / Neumark	122	45.49 / 78.75	53.69 / 55.70	0.168
		Female / Bertrand	18	45.38 / 78.55	60.75 / 62.87	0.083
		Female / Kline	38	45.36 / 78.47	55.49 / 52.14	0.241
		Female / Farber	12	45.48 / 78.87	58.41 / 60.51	0.111
		Female / Neumark	77	45.44 / 78.75	54.16 / 55.93	0.132
		Male / Bertrand	18	45.51 / 78.64	52.50 / 60.01	0.074
		Male / Kline	38	45.53 / 78.70	53.13 / 56.24	0.240
		Male / Neumark	45	45.58 / 78.76	52.88 / 55.30	0.229
Qwen3-14B	Race	Black / Bertrand	18	-8.48 / 17.36	57.96 / 60.93	0.062
		Black / Kline	38	-8.52 / 17.31	46.37 / 45.61	0.231
		White / Bertrand	18	-9.29 / 17.76	55.30 / 61.95	0.094
		White / Kline	38	-9.30 / 17.75	62.26 / 62.77	0.251
		White / Farber	12	-9.21 / 17.72	58.41 / 60.51	0.111
		White / Neumark	122	-9.20 / 17.63	53.69 / 55.70	0.168
		Female / Bertrand	18	-8.75 / 17.77	60.75 / 62.87	0.083
		Female / Kline	38	-8.70 / 17.68	55.49 / 52.14	0.241
		Female / Farber	12	-9.21 / 17.72	58.41 / 60.51	0.111
		Female / Neumark	77	-9.07 / 17.71	54.16 / 55.93	0.132
		Male / Bertrand	18	-9.02 / 17.35	52.50 / 60.01	0.074
		Male / Kline	38	-9.12 / 17.38	53.13 / 56.24	0.240
		Male / Neumark	45	-9.43 / 17.50	52.88 / 55.30	0.229
Qwen3.6-27B	Race	Black / Bertrand	18	5.26 / 8.10	57.96 / 60.93	0.062
		Black / Kline	38	5.20 / 8.07	46.37 / 45.61	0.231
		White / Bertrand	18	5.20 / 8.18	55.30 / 61.95	0.094
		White / Kline	38	5.22 / 8.22	62.26 / 62.77	0.251
		White / Farber	12	5.19 / 8.12	58.41 / 60.51	0.111
		White / Neumark	122	5.18 / 8.17	53.69 / 55.70	0.168
		Female / Bertrand	18	5.19 / 8.03	60.75 / 62.87	0.083
		Female / Kline	38	5.20 / 8.05	55.49 / 52.14	0.241
		Female / Farber	12	5.19 / 8.12	58.41 / 60.51	0.111

Tab. S.12 continued

Model	Axis	Group × Study	n	Model warmth/competence	Human warmth/competence	Human callback
		Female / Neumark	77	5.17 / 8.10	54.16 / 55.93	0.132
		Male / Bertrand	18	5.27 / 8.25	52.50 / 60.01	0.074
		Male / Kline	38	5.22 / 8.23	53.13 / 56.24	0.240
		Male / Neumark	45	5.21 / 8.29	52.88 / 55.30	0.229
Qwen3.6-35B-A3B	Race	Black / Bertrand	18	0.17 / 0.39	57.96 / 60.93	0.062
		Black / Kline	38	0.17 / 0.39	46.37 / 45.61	0.231
		White / Bertrand	18	0.19 / 0.40	55.30 / 61.95	0.094
		White / Kline	38	0.18 / 0.40	62.26 / 62.77	0.251
		White / Farber	12	0.19 / 0.40	58.41 / 60.51	0.111
		White / Neumark	122	0.18 / 0.40	53.69 / 55.70	0.168
		Female / Bertrand	18	0.18 / 0.39	60.75 / 62.87	0.083
		Female / Kline	38	0.18 / 0.40	55.49 / 52.14	0.241
		Female / Farber	12	0.19 / 0.40	58.41 / 60.51	0.111
		Female / Neumark	77	0.18 / 0.40	54.16 / 55.93	0.132
		Male / Bertrand	18	0.18 / 0.40	52.50 / 60.01	0.074
		Male / Kline	38	0.18 / 0.40	53.13 / 56.24	0.240
		Male / Neumark	45	0.17 / 0.39	52.88 / 55.30	0.229

Bootstrap Mediation, All Nine Models

Indirect effects for the path name group to probe score to callback margin, for every model, both groupings, and both probes. Fourteen of the thirty-six unadjusted 95% bootstrap intervals exclude zero. No multiplicity-adjusted inference is reported, so the coefficients are provided for exploratory comparison rather than as confirmed path selections.

Model	Grouping	Probe	a	b	IE	95% CI
Gemma-3-12B	Race (Black=1)	Warmth	-0.502	+0.171	-0.086	[-0.186, +0.019]
	Race (Black=1)	Competence	-0.501	+0.174	-0.087	[-0.188, +0.018]
	Gender (Female=1)	Warmth	+0.072	+0.236	+0.017	[-0.011, +0.056]
	Gender (Female=1)	Competence	+0.068	+0.240	+0.016	[-0.012, +0.055]
Gemma-3-27B	Race (Black=1)	Warmth	-0.391	-0.114	+0.045	[-0.008, +0.106]
	Race (Black=1)	Competence	-0.383	-0.108	+0.041	[-0.010, +0.102]
	Gender (Female=1)	Warmth	+0.093	-0.066	-0.006	[-0.027, +0.006]
	Gender (Female=1)	Competence	+0.096	-0.055	-0.005	[-0.025, +0.007]
Llama-3.1-8B	Race (Black=1)	Warmth	+0.519	+0.366	+0.190†	[+0.111, +0.292]
	Race (Black=1)	Competence	+0.218	+0.185	+0.040†	[+0.004, +0.090]
	Gender (Female=1)	Warmth	+0.027	-0.112	-0.003	[-0.022, +0.016]
	Gender (Female=1)	Competence	+0.208	-0.012	-0.003	[-0.035, +0.030]
Gemma-4-12B	Race (Black=1)	Warmth	-0.260	-0.109	+0.028	[-0.003, +0.069]
	Race (Black=1)	Competence	-0.515	+0.291	-0.150†	[-0.295, -0.019]
	Gender (Female=1)	Warmth	-0.327	-0.119	+0.039	[-0.004, +0.079]
	Gender (Female=1)	Competence	-0.032	-0.024	+0.001	[-0.015, +0.011]
Gemma-4-26B-A4B	Race (Black=1)	Warmth	+0.476	+0.173	+0.082	[-0.038, +0.194]
	Race (Black=1)	Competence	+0.387	+0.339	+0.131†	[+0.060, +0.231]
	Gender (Female=1)	Warmth	+0.213	+0.181	+0.038†	[+0.007, +0.078]
	Gender (Female=1)	Competence	-0.073	+0.386	-0.028	[-0.076, +0.023]
Gemma-4-31B	Race (Black=1)	Warmth	-0.372	+0.143	-0.053	[-0.164, +0.005]
	Race (Black=1)	Competence	-0.465	+0.494	-0.230†	[-0.483, -0.049]
	Gender (Female=1)	Warmth	-0.094	+0.088	-0.008	[-0.042, +0.002]
	Gender (Female=1)	Competence	+0.010	+0.186	+0.002	[-0.029, +0.026]
Qwen3-14B	Race (Black=1)	Warmth	+0.488	+0.165	+0.081†	[+0.013, +0.156]
	Race (Black=1)	Competence	-0.255	+0.517	-0.132†	[-0.217, -0.058]
	Gender (Female=1)	Warmth	+0.210	+0.014	+0.003	[-0.017, +0.024]
	Gender (Female=1)	Competence	+0.218	+0.347	+0.076†	[+0.029, +0.141]
Qwen3.6-27B	Race (Black=1)	Warmth	+0.070	+0.411	+0.029	[-0.029, +0.106]
	Race (Black=1)	Competence	-0.169	+0.290	-0.049†	[-0.103, -0.011]
	Gender (Female=1)	Warmth	-0.021	+0.415	-0.009	[-0.059, +0.047]
	Gender (Female=1)	Competence	-0.320	+0.383	-0.123†	[-0.206, -0.061]
Qwen3.6-35B-A3B	Race (Black=1)	Warmth	-0.228	+0.454	-0.103†	[-0.169, -0.046]
	Race (Black=1)	Competence	-0.097	+0.354	-0.034	[-0.080, +0.006]
	Gender (Female=1)	Warmth	+0.249	+0.257	+0.064†	[+0.027, +0.113]
	Gender (Female=1)	Competence	+0.124	+0.234	+0.029†	[+0.001, +0.067]

Table S.13: Bootstrap mediation of name-group callback disparities through warmth and competence probe scores, all nine models. Path: name group $\rightarrow$ probe score $\rightarrow$ callback margin, standardized coefficients, Baron–Kenny/Preacher–Hayes bootstrap (5,000 resamples, seed 20260527, n = 269, race subset n = 227). a is the coefficient of group on probe score; b is the coefficient of probe score on callback margin, controlling for group; IE = $a \times b$ is the indirect (mediated) effect. † marks a 95% bootstrap CI that excludes zero. 14 of 36 intervals exclude zero at this unadjusted threshold. No multiplicity-adjusted inference is reported, so all paths should be read as exploratory.

Group	n	Human warmth/competence (z)	Human callback
Black-Female	28	-0.24 / -0.66	0.175
Black-Male	28	-0.37 / -0.41	0.178
White-Female	117	+0.23 / +0.02	0.148
White-Male	73	+0.03 / +0.03	0.216

Model	B-F	B-M	W-F	W-M
Gemma-3-12B	-0.46/-0.47	-0.14/-0.14	+0.65/+0.64	+0.42/+0.41
Gemma-3-27B	-0.56/-0.52	-0.19/-0.15	+0.50/+0.51	+0.22/+0.23
Llama-3.1-8B	+0.62/+0.70	+0.47/-0.17	-0.31/-0.13	-0.39/-0.43
Gemma-4-12B	-0.50/-0.48	-0.09/-0.17	+0.02/+0.49	+0.71/+0.56
Gemma-4-26B-A4B	+1.02/+0.25	-0.03/-0.04	-0.28/-0.63	-0.64/-0.59
Gemma-4-31B	-0.34/-0.31	+0.13/+0.04	+0.19/+0.41	+0.45/+0.42
Qwen3-14B	+0.53/-0.11	+0.07/-0.87	-0.41/+0.17	-0.83/-0.12
Qwen3.6-27B	+0.15/-0.61	+0.13/-0.07	-0.13/-0.24	+0.20/+0.61
Qwen3.6-35B-A3B	-0.08/+0.07	-0.14/+0.10	+0.64/+0.39	+0.24/+0.28

Model margin	B-F	B-M	W-F	W-M
Gemma-3-12B	-0.183	-0.290	-0.169	-0.276
Gemma-3-27B	+1.621	+1.781	+1.019	+1.293
Llama-3.1-8B	-1.964	-2.049	-2.060	-2.096
Gemma-4-12B	+17.125	+17.221	+17.167	+17.124
Gemma-4-26B-A4B	+21.815	+21.158	+21.274	+20.672
Gemma-4-31B	+25.888	+25.787	+25.832	+25.285
Qwen3-14B	+1.817	+1.621	+1.844	+1.452
Qwen3.6-27B	+5.420	+5.362	+5.386	+5.226
Qwen3.6-35B-A3B	+3.112	+2.875	+3.193	+2.827

Table S.14: Crossed race × gender group levels. Human values are printed once because they are shared across models. The two model grids report warmth/competence z-scores and raw callback margins. B-F, B-M, W-F, and W-M denote Black-female, Black-male, White-female, and White-male name groups. Study-specific raw values are in Tab. S.15.

Crossed Race × Gender Disparity, All Nine Models

Race and gender are treated as separate marginal axes throughout the main text. Crossing them recovers interaction structure the marginal breakdown collapses. No result in this paper rests on the table below; it is reported because the gender gap is visibly not independent of race in most checkpoints, and that observation motivates the intersectional direction noted in Future Work. Cell sizes are 28 Black-female, 28 Black-male, 117 White-female, and 73 White-male names, and no interval estimates or significance tests were computed for the interaction terms.

Name-Level Warmth/Competence by Crossed Race × Gender

Tab. S.15 is the raw-value companion to Tab. S.14 above: race and gender crossed into four joint cells (Black-Female, Black-Male, White-Female, White-Male), recovering the interaction structure the marginal breakdown collapses. Within each cell, callback rates are further broken down by source study rather than blended together, for the same reason as above. Cell sizes range from a single-study group of 9 names (each Black cell under Bertrand alone) to 91 distinct names, 117 name-study observations, in White-Female overall, so the crossed cell means should be read as descriptive rather than as adequately powered group comparisons.

Table S.15: Name-level warmth/competence and callback margin by crossed race × gender group, broken down by source study, raw values. Applicant names are joined to Gallo et al. [11]'s published race/gender/callback labels by lowercase first name and matching study (`src/hiring_r4.py`); a name rated under more than one source study (e.g. Bertrand and Kline) contributes one row per study rather than being collapsed to a single value, since callback rates differ meaningfully by study. Model warmth/competence are raw projections and are not comparable in magnitude across models or against the 0-100 human warmth/competence scale; see Tab. S.14 for the standardized (z-score), study-collapsed comparison.

Model	Race × Gender	Study	n	Model warmth/competence	Human warmth/competence	Human callback	Model margin
Gemma-3-12B	Black-Female	Bertrand	9	37660.74 / 40372.96	60.95 / 61.19	0.065	-0.181
		Kline	19	37546.01 / 40249.57	46.16 / 43.04	0.227	-0.184
	Black-Male	Bertrand	9	37842.95 / 40574.00	54.97 / 60.66	0.059	-0.278
		Kline	19	37862.93 / 40599.59	46.58 / 48.18	0.234	-0.296
	White-Female	Bertrand	9	38623.69 / 41404.47	60.55 / 64.55	0.100	-0.208
		Kline	19	38636.22 / 41418.95	64.83 / 61.24	0.255	-0.184
		Farber	12	38682.20 / 41473.86	58.41 / 60.51	0.111	-0.156
		Neumark	77	38474.98 / 41250.92	54.16 / 55.93	0.132	-0.162
	White-Male	Bertrand	9	38401.65 / 41170.59	50.04 / 59.36	0.089	-0.222
		Kline	19	38431.41 / 41209.65	59.68 / 64.30	0.247	-0.250
		Neumark	45	38284.64 / 41048.87	52.88 / 55.30	0.229	-0.297
Gemma-3-27B	Black-Female	Bertrand	9	26107.43 / 25164.89	60.95 / 61.19	0.065	1.514
		Kline	19	25997.99 / 25053.18	46.16 / 43.04	0.227	1.671
	Black-Male	Bertrand	9	26274.15 / 25309.69	54.97 / 60.66	0.059	1.778
		Kline	19	26264.25 / 25303.71	46.58 / 48.18	0.234	1.783
	White-Female	Bertrand	9	26846.20 / 25851.30	60.55 / 64.55	0.100	1.125
		Kline	19	26817.69 / 25821.04	64.83 / 61.24	0.255	1.059
		Farber	12	26786.33 / 25790.22	58.41 / 60.51	0.111	0.896
		Neumark	77	26626.71 / 25638.55	54.16 / 55.93	0.132	1.016
	White-Male	Bertrand	9	26662.64 / 25668.05	50.04 / 59.36	0.089	1.333
		Kline	19	26652.83 / 25660.59	59.68 / 64.30	0.247	1.336
		Neumark	45	26436.70 / 25453.09	52.88 / 55.30	0.229	1.267
Llama-3.1-8B	Black-Female	Bertrand	9	-1.26 / -0.15	60.95 / 61.19	0.065	-1.979
		Kline	19	-1.25 / -0.15	46.16 / 43.04	0.227	-1.957
	Black-Male	Bertrand	9	-1.27 / -0.18	54.97 / 60.66	0.059	-2.069
		Kline	19	-1.27 / -0.17	46.58 / 48.18	0.234	-2.039
	White-Female	Bertrand	9	-1.35 / -0.17	60.55 / 64.55	0.100	-2.021
		Kline	19	-1.35 / -0.17	64.83 / 61.24	0.255	-2.033
		Farber	12	-1.34 / -0.17	58.41 / 60.51	0.111	-2.062
		Neumark	77	-1.33 / -0.17	54.16 / 55.93	0.132	-2.071
	White-Male	Bertrand	9	-1.35 / -0.18	50.04 / 59.36	0.089	-2.076
		Kline	19	-1.36 / -0.18	59.68 / 64.30	0.247	-2.079
		Neumark	45	-1.34 / -0.17	52.88 / 55.30	0.229	-2.107
Gemma-4-12B	Black-Female	Bertrand	9	12.19 / 18.40	60.95 / 61.19	0.065	17.121
		Kline	19	12.21 / 18.41	46.16 / 43.04	0.227	17.127
	Black-Male	Bertrand	9	12.22 / 18.51	54.97 / 60.66	0.059	17.266
		Kline	19	12.26 / 18.63	46.58 / 48.18	0.234	17.199
	White-Female	Bertrand	9	12.30 / 19.06	60.55 / 64.55	0.100	17.222
		Kline	19	12.29 / 19.07	64.83 / 61.24	0.255	17.161
		Farber	12	12.26 / 19.06	58.41 / 60.51	0.111	17.186
		Neumark	77	12.25 / 18.96	54.16 / 55.93	0.132	17.159
	White-Male	Bertrand	9	12.31 / 19.06	50.04 / 59.36	0.089	17.073
		Kline	19	12.32 / 19.10	59.68 / 64.30	0.247	17.128
		Neumark	45	12.36 / 19.01	52.88 / 55.30	0.229	17.132
Gemma-4-26B-A4B	Black-Female	Bertrand	9	1.14 / 5.56	60.95 / 61.19	0.065	21.840
		Kline	19	1.18 / 5.58	46.16 / 43.04	0.227	21.803
	Black-Male	Bertrand	9	0.98 / 5.49	54.97 / 60.66	0.059	21.160
		Kline	19	0.96 / 5.54	46.58 / 48.18	0.234	21.158
	White-Female	Bertrand	9	0.92 / 5.40	60.55 / 64.55	0.100	21.271
		Kline	19	0.92 / 5.40	64.83 / 61.24	0.255	21.280
		Farber	12	0.88 / 5.42	58.41 / 60.51	0.111	21.229
		Neumark	77	0.92 / 5.44	54.16 / 55.93	0.132	21.280
	White-Male	Bertrand	9	0.84 / 5.48	50.04 / 59.36	0.089	20.840
		Kline	19	0.81 / 5.45	59.68 / 64.30	0.247	20.793
		Neumark	45	0.86 / 5.43	52.88 / 55.30	0.229	20.587
Gemma-4-31B	Black-Female	Bertrand	9	45.20 / 78.05	60.95 / 61.19	0.065	25.889
		Kline	19	45.21 / 77.98	46.16 / 43.04	0.227	25.888

Tab. S.15 continued

Model	Race × Gender	Study	n	Model warmth/competence	Human warmth/competence	Human callback	Model margin
	Black-Male	Bertrand	9	45.41 / 78.30	54.97 / 60.66	0.059	25.781
		Kline	19	45.45 / 78.44	46.58 / 48.18	0.234	25.789
	White-Female	Bertrand	9	45.56 / 79.05	60.55 / 64.55	0.100	25.924
		Kline	19	45.51 / 78.96	64.83 / 61.24	0.255	25.888
		Farber	12	45.48 / 78.87	58.41 / 60.51	0.111	25.888
		Neumark	77	45.44 / 78.75	54.16 / 55.93	0.132	25.798
	White-Male	Bertrand	9	45.61 / 78.98	50.04 / 59.36	0.089	25.694
		Kline	19	45.60 / 78.96	59.68 / 64.30	0.247	25.708
		Neumark	45	45.58 / 78.76	52.88 / 55.30	0.229	25.024
Qwen3-14B	Black-Female	Bertrand	9	-8.39 / 17.61	60.95 / 61.19	0.065	1.806
		Kline	19	-8.27 / 17.54	46.16 / 43.04	0.227	1.822
	Black-Male	Bertrand	9	-8.57 / 17.11	54.97 / 60.66	0.059	1.667
		Kline	19	-8.76 / 17.07	46.58 / 48.18	0.234	1.599
	White-Female	Bertrand	9	-9.11 / 17.92	60.55 / 64.55	0.100	1.847
		Kline	19	-9.13 / 17.81	64.83 / 61.24	0.255	1.829
		Farber	12	-9.21 / 17.72	58.41 / 60.51	0.111	1.917
		Neumark	77	-9.07 / 17.71	54.16 / 55.93	0.132	1.836
	White-Male	Bertrand	9	-9.47 / 17.59	50.04 / 59.36	0.089	1.514
		Kline	19	-9.48 / 17.68	59.68 / 64.30	0.247	1.526
		Neumark	45	-9.43 / 17.50	52.88 / 55.30	0.229	1.408
Qwen3.6-27B	Black-Female	Bertrand	9	5.22 / 8.00	60.95 / 61.19	0.065	5.431
		Kline	19	5.22 / 8.02	46.16 / 43.04	0.227	5.414
	Black-Male	Bertrand	9	5.30 / 8.20	54.97 / 60.66	0.059	5.361
		Kline	19	5.18 / 8.11	46.58 / 48.18	0.234	5.362
	White-Female	Bertrand	9	5.16 / 8.06	60.55 / 64.55	0.100	5.403
		Kline	19	5.18 / 8.09	64.83 / 61.24	0.255	5.414
		Farber	12	5.19 / 8.12	58.41 / 60.51	0.111	5.458
		Neumark	77	5.17 / 8.10	54.16 / 55.93	0.132	5.365
	White-Male	Bertrand	9	5.23 / 8.30	50.04 / 59.36	0.089	5.264
		Kline	19	5.27 / 8.34	59.68 / 64.30	0.247	5.296
		Neumark	45	5.21 / 8.29	52.88 / 55.30	0.229	5.189
Qwen3.6-35B-A3B	Black-Female	Bertrand	9	0.17 / 0.39	60.95 / 61.19	0.065	3.167
		Kline	19	0.17 / 0.39	46.16 / 43.04	0.227	3.086
	Black-Male	Bertrand	9	0.17 / 0.39	54.97 / 60.66	0.059	2.903
		Kline	19	0.17 / 0.39	46.58 / 48.18	0.234	2.862
	White-Female	Bertrand	9	0.19 / 0.40	60.55 / 64.55	0.100	3.250
		Kline	19	0.19 / 0.40	64.83 / 61.24	0.255	3.237
		Farber	12	0.19 / 0.40	58.41 / 60.51	0.111	3.229
		Neumark	77	0.18 / 0.40	54.16 / 55.93	0.132	3.170
	White-Male	Bertrand	9	0.18 / 0.41	50.04 / 59.36	0.089	2.847
		Kline	19	0.18 / 0.40	59.68 / 64.30	0.247	2.862
		Neumark	45	0.17 / 0.39	52.88 / 55.30	0.229	2.808